Biology

TARGET : Pre-Medical

Diwali Assignment

(MEDIUM : ENGLISH)

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PREFACE

Dear students,

This **Diwali Assignment Question Bank** for **Students** of has been thoughtfully created to provide students with a comprehensive set of questions aimed at reinforcing core concepts during the Diwali festive break. It balances revision with active engagement, ensuring effective learning even during holidays. Our experience shows that if students utilize the Diwali break effectively, their academic performance improves significantly after Diwali. We suggest that students maintain a healthy balance between study and relaxation – using the morning and night hours for focused study, and keeping time aside for relaxation and refreshment. For this assignment, we advised you that you have to practice by setting with alarm clock, read once your related chapter of NCERT and solve the questions carefully.

This assignment will provide you following core implementation in your study and a perfect revision plan.

1. **Strong Your Topics :**

   We are providing you an assignment of important questions for the Diwali holidays. Solve the questions from the topics that have been covered in your class so far. The remaining questions of uncovered syllabus can be completed after the Diwali break.

   All these questions have been specially prepared by the Allen Kota faculty to help you improve your NEET preparation. We expect you to solve them with complete dedication, sincerity, and concentration.

2. **Focus on Conceptual Clarity:**

   The questions are designed to test **depth of understanding**, not just rote memorization. Special emphasis has been laid on **conceptual application** of each chapter and **logical thinking**.

3. **Betterment Strategy for Students:**
   - This assignment include move than Approx 400 questions so student can solve approx 50 questions per day is Diwali vacations & feel free and energetic during holidays.
   - Encourages **self-assessment** and **reflection** on strengths and weaknesses.
   - Enhances **exam-readiness** by simulating actual question formats found in previous NEET competitive exams.
   - Provides an opportunity to **stay academically active** during festive downtime in a stress-free manner.
   - Builds **confidence** in handling diverse question patterns.

4. **Student-Centric Approach:**
   - Offers **progressive difficulty leaves** to cater to all types of learners.
   - Allows flexibility-students can choose to attempt questions based on topics they find challenging.

5. **Allen Kota Biology department's Recommendation:**
   - Students are advised to attempt the question bank **honestly and independently**, using it as a tool to **diagnose learning gaps** and **sharpen exam techniques**. The effort invested during this time will yield results in upcoming NEET examination.

HAPPY DIWALI

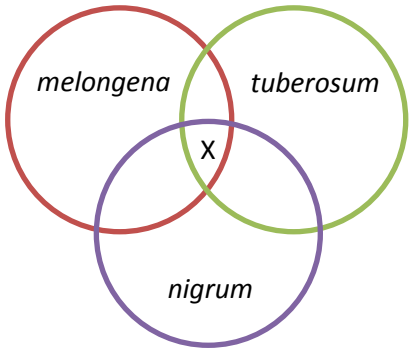
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BIOLOGY DIWALI ASSIGNMENT

S.No.	Contents	Page
1.	Plant Diversity (The living world, Biological classification and Plant kingdom)	1
2.	Animal Kingdom	6
3.	Morphology of Flowering Plants	8
4.	Anatomy of Flowering Plants	11
5.	Structural organisation of animals (Animal tissues)	14
6.	Structural organisation of animals (Cockroach and Frog)	16
7.	Cell - The unit of life	22
8.	Biomolecules	25
9.	Cell cycle & Cell division	28
10.	Photosynthesis in Higher Plants	31
11.	Respiration in Plants	36
12.	Plant Growth and Development	40
13.	Breathing and Exchange of Gases	44
14.	Body Fluids and Circulation	46
15.	Excretory products & their elimination	48
16.	Movement and Locomotion (Muscles & skeletal system)	49
17.	Neural control & Co-ordination	52
18.	Chemical Coordination and Integration	53
19.	Sexual Reproduction in Flowering Plants	54
20.	Human Reproduction	58
21.	Reproductive Health	60
22.	Principles of Inheritance and Variation	62
23.	Molecular Basis of Inheritance	66
24.	Evolution	70
25.	Human Health & Disease	72
26.	Biotechnology : Principle and Processes	75
27.	Biotechnology and its application and Microbes in Human Welfare	78
28.	Ecology	83

PLANT DIVERSITY
(THE LIVING WORLD, BIOLOGICAL CLASSIFICATION AND PLANT KINGDOM)

1. Given below is a Venn diagram. Analyse the given diagram and choose the incorrect option from options below.



(1) X here denotes the genus
(2) *tuberosum* and *nigrum* are the specific epithets of genus-*Solanum*
(3) *S. tuberosum* and *S. melongena* share morphologically similar characters and thus are same species of different genus.
(4) One species can be distinguished from other closely related one based on distinct differences based on morphology.

2. **Assertion (A) :** The problem of classification becomes more complex on moving from lower to higher taxa.
Reason (R) : Higher the category, less are the number of characters shared.
(1) If both A and R are true and R is the correct explanation of A.
(2) If both A and R are true but R is not the correct explanation of A.
(3) If A is true, but R is false.
(4) If A is false, but R is true.

3. Select the incorrect statement w.r.t. heterocysts.
(1) These are present in BGA
(2) These help in N_2 fixation
(3) *Nostoc* and *Rhizopus* are some organisms which have these cells
(4) Organisms having these cells have chlorophyll-a pigment similar to green plants.

4. Which of the following statement is incorrect regarding the class-Basidiomycetes?
(1) They usually grow in soil, logs, tree stumps and in living plant bodies.
(2) Their commonly known forms are mushrooms, bracket fungi or puffballs.
(3) These are the only class of fungi which are not parasitic in nature.
(4) They lack sex organs, but plasmogamy takes place by the fusion of two vegetative cells.

5. Which of the following are the common parasites of class-Basidomyctes?
(1) *Ustilago* and *Puccina*
(2) *Agaricus* and *Trichorderma*
(3) *Alternaria* and *Colletotrichum*
(4) *Colletotrichum* and *Puccinia*

6. **Assertion (A) :** Viruses cause diseases and replicate when they are inside the host cell.

Reason (R) : Viruses do not replicate outside the host, but they survive in environment.

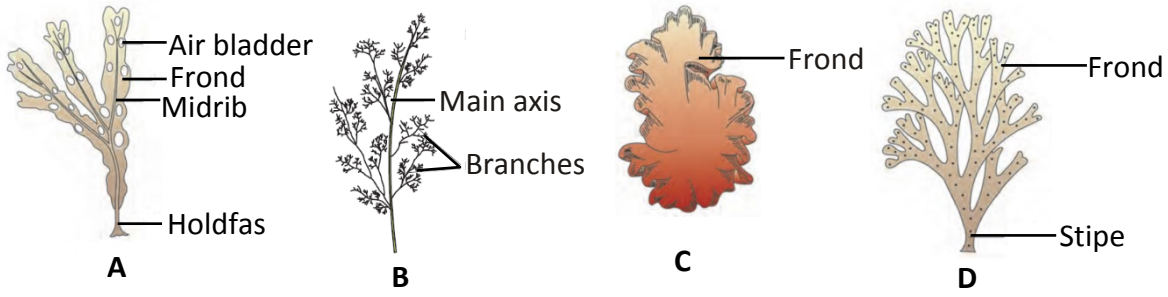
(1) If both A and R are true and R is the correct explanation of A.

(2) If both A and R are true but R is not the correct explanation of A.

(3) If A is true, but R is false.

(4) If A is false, but R is true.

7. Identify the given figures of algae and select the correct option.



(1) A-*Volvox,* B-*Chlamydomonas,* C-*Chara,* D-*Porphyra*

(2) A-*Fucus*, B-*Polysiphonia*, C-*Porphyra*, D-*Dictyota*

(3) A-*Fucus*, B-*Dictyota*, C-*Porphyra*, D-*Polysiphonia*

(4) A-*Dictyota*, B-*Porphyra*, C-*Fucus*, D-*Polysiphonia*

8. Oogamy means

(1) fusion between female and male gametes of similar size

(2) fusion between one large motile female gamete and a smaller non-motile male gamete

(3) fusion between one large non-motile female gamete and a smaller motile male gamete

(4) fusion between one smaller female gamete and a large motile male gamete

9. **Assertion (A) :** In pteriodophytes, zygote produces a multicellular sporophyte.

Reason (R) : The zygote develops into young embryos within the sporophyte in most of pteridophytes.

(1) If both A and R are true and R is the correct explanation of A.

(2) If both A and R are true but R is not the correct explanation of A.

(3) If A is true, but R is false.

(4) If A is false, but R is true.

10. **Statement–I :** In conifers, the needle like leaves reduce the surface area.

Statement–II : Leaves in gymnosperms are adapted to withstand extremes of temperature, humidity and wind.

(1) Both statement I and statement II are correct

(2) Statement I is correct, but statement II is incorrect

(3) Statement I is incorrect, but statement II is correct

(4) Both statement I and statement II are incorrect

11. Read the statements carefully and select the correct statement –

(1) Prokaryotic cells are represented by fungi and bacteria.

(2) Prokaryotic cell have a cell wall (Cellulose)and cell membrane.

(3) In addition to the genomic DNA, bacteria have smaller double stranded, circular DNA called as "Plasmid".

(4) Membrane bound organelle are presented in bacterial cytoplasm.

12. How many organisms in the list given below are autotrophs ?
Lactobacillus, Nostoc, Chara, Nitrosomonas, Nitrobacter, Streptomyces, Saccharomyces, Trypanosoma, Porphyra, Wolfia
(1) Six (2) Three (3) Four (4) Five

13. Conifers differ from grasses in the
(1) absence of pollen tubes
(2) formation of endosperm before fertilization
(3) production of seeds from ovules
(4) lack of xylem tracheids

14. **Assertion :** Heterosporous Pteridophyte represents the event that is precursor to the seed habit.
Reason : Development of the zygote into young embryo take place within the female gametophyte and female gametophyte is retained on the parent sporophyte for variable periods in heterosporous pterigophytes.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

15. Select the correct statement :
(1) Genus is a group of related species which has less characters in common in comparison to species of other genera
(2) Families are characterised on the basis of only reproductive features of plant species
(3) Each genus has only one specific epithet representing different organisms.
(4) Classes with few similar characters are assigned to a higher category called division.

16. Which of the following is true for nomenclature?
(1) Naming of all living organisms is based on their features.
(2) Naming of an organism by the same name all over the world.
(3) Naming of living organisms in a manner that is becomes easy to pronounce.
(4) Naming of an organism, so that it causes less confusion.

17. Arrange the following in ascending order of common characters :-
I. Family II. Genus III. Class IV. Species
(1) Class < Family < Genus < Species
(2) Family < Class < Genus < Spcies
(3) Species < Genus < Family < Class
(4) Class < Genus < Species < Family

18. Some of the features of Archaebacteria are given below.
I. Some are found in the gut of ruminants.
II. Some are able to survive in halophilic conditions.
III. All are autotrophic organisms.
IV. Fix atmospheric nitrogen.
V. Produce biogas from dung.
How many features are correct w.r.t. Archaebacteria?
(1) Two (2) Three (3) Four (4) Five

19. Which of the following bacteria form blooms in polluted water bodies?
(1) Cyanobacteria
(2) Chemosynthetic autotrophs
(3) Heterotrophic bacteria
(4) Archaebacteria

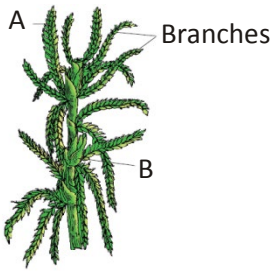
20. Members of Ascomycetes possess which of the following characteristics?
I. Saprophytic
II. Decomposers
III. Parasitic or coprophilous
IV. Aseptate mycelium
Choose the correct options.
(1) I, III and IV
(2) II, III and IV
(3) I, II and III
(4) I, II and IV

21. **Assertion (A) :** In fungi, sexual apparatus decreases in complexity from lower to higher form.
Reason (R) : In plants, sexual apparatus increases in complexity from simple to higher forms.
(1) If both A and R are true and R is the correct explanation of A.
(2) If both A and R are true but R is not the correct explanation of A.
(3) If A is true, but R is false.
(4) If A is false, but R is true.

22. The phylogenetic classification systems is based on evolutionary relationships between the various organisms, it assumes that
(1) organisms belonging to the different taxa have a common ancestor.
(2) plants belonging to the same taxa have a common reproductive strategies.
(3) organisms belonging to the different genera have a different ancestor.
(4) organisms belonging to the same taxa have a common ancestor.

23. Which of the following is an incorrect representation of algae?
(1) They are chlorophyll bearing autotrophs
(2) They are simple thalloid forms that are aquatic
(3) Some algae occur in association with fungi and animals
(4) Zoospores are non-flagellated spores that aid in asexual reproduction

24. A and B in given figure represent



(1) A-Gametophyte branch, B-Sporophyte branch
(2) A-Antheridial branch, B-Archegonial branch
(3) A-Archegonial branch, B-Antheridial branch
(4) A-Sporophyte branch, B-Gametophyte branch

25. **Assertion (A) :** In gymnosperms, ovules are enclosed by an ovary wall and remain exposed after fertilisation.
Reason (R) : Seeds are naked and are developed after fertilisation.
(1) If both A and R are true and R is the correct explanation of A.
(2) If both A and R are true but R is not the correct explanation of A.
(3) If A is true, but R is false.
(4) If A is false, but R is true.

26. Fill up the blanks in the following statements with the appropriate options.
I. Heterocyst is a specialised cyanobacterial cell that carries out ....(A)....
II. Cyanobacteria have ....(B).... similar to greenplants.
III. Cyanobacteria often form ....(C).... in polluted water bodies.
IV. Some species of ....(D).... secrete toxins, which inhibit the growth of mosquito larva in water.

	(A)	(B)	(C)	(D)
1	Photosynthesis	Carotenoids	Death factor	Anabaena
2	Nitrogen fixation	Chlorophyll-a	Water blooms	Oscillatoria
3	Reproduction	Chlorophyll-b	Water blooms	Nostoc
4	Metabolism	Chlorophyll-a	trichome	Oscillatoria

27. Read the following points

(a) Two flagella

(b) Photosynthetic ability

(c) Aquatic habitat

(d) Unicellular eukaryotes

above points are related with :-

(1) Chrysophytes and dinoflagellates (2) Dinoflagellates and slime moulds

(3) Dinoflagellates and euglenoids (4) Chrysophytes and euglenoids

28. Identify the following figures and choose the option which gives correct description about them :-



A B C D

(1) B – *Chara* → Sexual organs are multicellular and jacketed

(2) A – *Cycas* → Male and female cones are borne on same tree

(3) D – TMV → Genetic material is ds DNA.

(4) C – *Euglena* → Shows mixotrophic nutrition.

29. The system of classification given by Carolus Linnaeus i.e. sexual system is an artificial system as into account:

(1) Anatomical characters were taken into account.

(2) Both physiological and morphological characters were taken into account.

(3) All possible characters of organisms were considered.

(4) Few morphological characters were taken into account.

30. Arrange the following taxa to form the correct sequence of classification of housefly:

(I) Diptera (II) Arthropoda

(III) Insecta (IV) Muscidae

(1) I, III, IV, II (2) IV, II, I, III (3) III, II, IV, I (4) II, III, I, IV

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**1**	**3**	**3**	**1**	**3**	**2**	**3**	**3**	**1**	**3**	**1**	**2**	**1**	**4**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**1**	**2**	**1**	**3**	**2**	**4**	**4**	**2**	**4**	**2**	**3**	**1**	**4**	**4**

ANIMAL KINGDOM

1. A vertebrate which doesn't possess jaws :

(1) *Balanoglossus* (2) *Petromyzon* (3) *Salamandra* (4) *Scoliodon*

2. Tube with in tube body plan (digestive system) is absent in :

(1) *Taenia* (2) *Ascaris* (3) Earthworms (4) Cockroach

3. Which one is not the character of Osteichthyes?

(1) Terminal mouth (2) Gills with operculum

(3) Body covered by placoid scales (4) Air bladder regulate buoyancy.

4. Which pair of animal has radial symmetry ?

(1) Sponge & Hydra (2) Hydra & Flatworm (3) Hydra & Echinus (4) Ascaris & Sycon

5. Which exclusive character of mammalia does not have any exception ?

(1) Body covered by hair (2) Ear pinna present

(3) Presence of mammary gland (4) Viviparous

6. Find the incorrect match :-

(1) Ascaris $\rightarrow$ Renett Cell (2) Limulus $\rightarrow$ Living fossil

(3) Locusta $\rightarrow$ Lac insect (4) Ancylostoma $\rightarrow$ Hook worm

7. Some important features are mentioned in the box below :

Oviparous only, Internal fertilization only, cold-blooded, placoid scales, mesonephric kidneys, streamlined body How many of the features are common both to chondrichthyes and osteichthyes ?

(1) Four (2) Three (3) Two (4) Five

8. Which of the following is a matching set of a phylum and its three example?

(1) Porifera : Spongilla, Euplectella,Pennatula

(2) Cnidaria : Bonellia, Physalia, Aurelia

(3) Platyhelminthes : Planaria, Schistosoma,Enterobius

(4) Mollusca : Loligo, Aplysia, Octopus

9. Tympanum represents ear in :-

(1) Amphibia (2) Reptile (3) Birds (4) Both (1) & (2)

10. Which member of amphibia is limbless ?

(1) Bufo (2) Hyla (3) Salamander (4) Ichthyophis

11. Which of the following character is not related with chordates ?

(1) Ventral nerve cord (2) Ventral heart

(3) Dorsal notochord (4) Post anal tail

12. The characteristic of birds which makes them different from reptiles:

(1) Homoeothermic nature (2) Post anal tail

(3) Vertebral column (4) Viviparity

13. True segmentation of body is first observed in the:-

(1) Platyhelminthes (2) Aschelminthes (3) Annelida (4) Arthropoda

14. In sponges, spongocoel is lined by :-

(1) Choanocytes (2) Amoebocytes (3) Pinacocytes (4) Archaeocytes

15. Worm like marine animals are present in :-

(1) Echinodermata (2) Mollusca (3) Ctenophora (4) Hemichordata

16. All of the following phyla are acoelomates except :-

(1) Porifera (2) Cnidaria (3) Ctenophora (4) Chordata

17. Find the incorrect one related to Pleurobrachia?

(1) Comb plates (2) Bioluminescence

(3) External fertilisation (4) Triploblastic

18. Which one of the following pairs of animals are similar to each other pertaining to the feature stated against them ?

(1) Ascaris and Ancylostoma-Metamericsegmentation

(2) Sea-horse and flying fish-Cold blooded(Poikilothermal)

(3) Pteropus and Ornithorhyncus-Viviparity

(4) Garden lizard and crocodile-Threechambered heart

19. Which of the following is pseudometamericsegmented animal ?

(1) Ascaris (2) Fasciola (3) Wuchereria (4) Tapeworm

20. Which of the following is correct for amphibia without exception.

(1) Cutaneous respiration (2) External fertilization

(3) Two pairs of limbs (4) Tympanum represents external ear

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**1**	**3**	**3**	**3**	**3**	**2**	**4**	**4**	**4**	**1**	**1**	**3**	**1**	**4**
Que.	16	17	18	19	20										
Ans.	**4**	**4**	**2**	**4**	**1**										

MORPHOLOGY OF FLOWERING PLANTS

1. Identify the correct set of statements :

(a) Prop roots are the hanging structures that support banyan tree.

(b) Stem is fleshy and cylindrical in *Euphorbia* and modified to perform the function of leaves.

(c) Leaves originate from shoot apical meristem and are arranged in acropetal order.

(d) The arrangement of vein and veinlets in the lamina of leaf is termed as venation.

(e) Axillary buds form spirally coiled tendrils in *Citrus* and *Bougainvillea*.

Choose the correct answer from the option given below :

(1) b, c, d and e only (2) a, b, c and d only (3) a, c, d and e only (4) a, b and d only

2. Match the column-I with column-II and choose the correct answer on the basis of codes given in column-I and column-II :

	Column-I		**Column-II**
I	Runner	a.	*Eichhornia*
II	Stolon	b.	Pineapple
III	Sucker	c.	Jasmine
IV	Offset	d.	Doob grass

(1) I-a, II-b, III-c, IV-d (2) I-b, II-a, III-d, IV-c (3) I-d, II-c, III-b, IV-a (4) I-d, II-b, III-a, IV-c

3. In how many of the following plant, actinomorphic symmetry is present in flower?

Gulmohar, Chilli, Tomato, Bean, Canna, Gram, Arhar, Mustard, Cassia

(1) Three (2) Four (3) Two (4) Five

4. (a) Leaves are arranged in acropetal or basipetal manner.

(b) Leaves originate from shoot apical meristem.

(c) Leaves develops at node and internodes.

Choose the correct statements :-

(1) a, b, c (2) a, b (3) b, c (4) only b

5. Replum is present in the ovary of flower of :

(1) Sunflower (2) Pea (3) Lemon (4) Mustard

6. A plant have following characters :-

Pulvinate leaf base, petals having vexillary aestivation, stamen-10, diadelphous condition, gynoecium-monocarpellary unilocular ovary with many ovules the plant belongs to :-

(1) Fabaceae (2) Solanaceae (3) Cruciferae (4) Malvaceae

7. Find the correct statements :-

(a) The dot above the floral diagram represents the mother axis

(b) The bracket represents the "cohesion" in floral formula

(c) Floral formula can show both cohesion and adhesion of stamens

(1) a, b and c (2) a and c only (3) b and c only (4) a and b only

8. Inferior ovary is found in -

(1) Malvaceae (2) Fabaceae (3) Cruciferae (4) Compositae

9. Trimerous flowers with superior ovary are present in :-

(1) Cotton (2) Wheat (3) Sunflower (4) China rose

10. **Statement-I :** Persistent calyx in the form of pappus is found in Compositae family.

Statement-II : Versatile stamens and trimerous flowers are characteristic feature of Poaceae family.

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Statement-I and statement-II both are correct

(4) Statement-I and statement-II both are incorrect

11. The lateral branches originate from the underground part of main stem, grow horizontally below the soil and then come out oblique upward giving rise to leafy shoots in all of the following except :-

(1) Banana (2) *Chrysanthemum* (3) Jasmine (4) Pineapple

12. **Statement 1:-** Flower is modified shoot where in shoot apical meristem changes to floral meristem

Statement 2 :- The floral apex produces different floral appendages laterally at successive internodes instead of leaves.

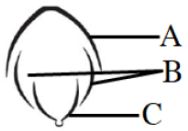
(1) Statement 1 is true while statement 2 is false

(2) Both statement 1 and 2 are false

(3) Both statements 1 and 2 is true

(4) Statement 1 is false while statement 2 is true

13. Identify A, B and C in given diagram :



(1) A = Standard, B = Keel, C = Wings

(2) A = Wing, B = Standard, C = Keel

(3) A = Standard, B = Wings, C = Keel

(4) A = Wings, B = Keel, C = Standard

14. Find out the correct option from the following with respect to cohesion of stamen :

	Monoadelphous stamen	**Diadelphous stamen**	**Polyadelphous stamen**	**Syngenesious stamen**
(1)	China rose	Pea	*Citrus*	Sunflower
(2)	Pea	Sunflower	China rose	*Citrus*
(3)	Sunflower	*Citrus*	China rose	Pea
(4)	*Citrus*	China rose	Pea	Sunflower

15. **Statement-I :** Position of gynoecium is not represented by floral diagram.

Statement-II : Aestivation can be represented by floral diagram.

(1) Statement I and II true (2) Statement I true and statement II false

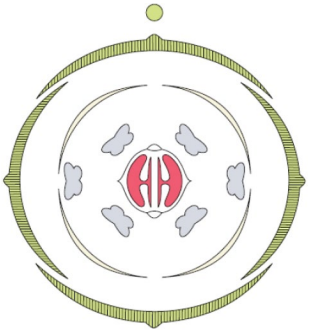
(3) Both statement I and II false (4) Statement II true and statement I false

16. Ray florets of sunflower have :

(1) Half inferior ovary (2) Superior ovary

(3) Inferior ovary (4) Hypogynous flower

17. Which of the floral formula is correct for given floral diagram of mustard ?



(1) ⊕ ♂♀ $K_{(5)}C_{1+2+(2)}A_{(9+1)}\underline{G}_{(2)}$ (2) % ♂♀ $K_{(5)}C_{1+2+(2)}A_{(9+1)}\underline{G}_{(2)}$

(3) % ♂♀ $K_{(2+2)}\,C_4A_{2+4}\underline{G}_{(2)}$ (4) ⊕ ♂♀ $K_{2+2}C_4A_{2+4}\underline{G}_{(2)}$

18. Study the statement carefully and give the answer :-

(A) Belladona and ashwagandha are the medicinal plants belong to Malvaceae family

(B) Lupin produce very beautiful flowers and belong to Fabaceae family

(C) Mustard is a member of family Brassicaceae

(D) *Aloe vera* is a member of family Fabaceae, used as medicinal plant.

How many statements are correct from these.

(1) Two (2) Three (3) Four (4) Five

19. Which of the following statement is correct :-

(1) In tomato fruit is a capsule (2) Seeds of orchids have oily endosperm

(3) Placentation in *Dianthus* is free central (4) Didynamous condition is found in pea

20. **Assertion (A) :** Staminal tube is present in Malvaceae.

Reason (R) : Ovary and styles are fused but stigmas remain free.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is NOT the correct explanation of (A)

(3) Both (A) and (R) are false.

(4) (A) is true but (R) is false.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**3**	**1**	**4**	**4**	**1**	**1**	**4**	**2**	**3**	**3**	**1**	**3**	**1**	**1**
Que.	16	17	18	19	20										
Ans.	**3**	**4**	**1**	**3**	**2**										

ANATOMY OF FLOWERING PLANTS

1. Monocot root, differs from monocot stem in having :-
(1) Open vascular bundles (2) Scattered vascular bundles
(3) Absenceq well developed pith (4) Radially arranged vascular bundles

2. Match the column I, II and III.

I		II		III	
A	Dicot leaf	a	Mesophyll tissue	(i)	Palisade parenchyma towards adaxial
B	Monocot leaf	b	Spongy and palisade parenchyma	(ii)	Mesophyll are not differentiated into palisade and spongy parenchyma
C	Dicot stem	c	Closed vascular bundle	(iii)	Starch sheath is present.
D	Monocot stem	d	Open vascular bundle	(iv)	Vascular bundle are scattered

Options :-
(1) A–a–ii, B–b–i, C–c–i, D–c–ii (2) A–b–i, B–a–ii, C–d–iii, D–c–iv
(3) A–c–iii, B–d–iii, C–b–ii, D–a–i (4) A–d–iv, B–c–iv, C–a–iv, D–b–iii

3. Phellam, Phellogen and phelloderm are collectively called as :-
(1) Cork cambium (2) Bark (3) Periderm (4) Secondary cortex

4. Arrange the following events of secondary growth in dicot stem incorrect sequence :-
(a) Formation of phellogen (b) Formation of fascicular cambium
(c) Formation of bark (d) Formation of wood
(1) a, d, b, c (2) b, d, a, c (3) b, c, a, d (4) a, b, d, c

5. Usually exarch and more than six (polyarch) xylem bandles are present in :-
(1) Monocot root (2) Dicot root (3) Dicot stem (4) Monocot leaf

6. Which of the following is not the part of epidermal tissue system ?
(1) Epidermis (2) Pericycle (3) Stomata (4) Trichomes

7. How many in the list given below are the part of ground tissue system.
Epidermis, Root hairs, Xylem, Phloem, Cortex, Pericycle, Pith
(1) One (2) Two (3) Three (4) Four

8. Read the following statements (A to D)
(A) The root hairs are unicellular elongations of epidermal cells.
(B) The root hairs help in preventing water loss due to transpiration.
(C) The trichomes in the shoot system are usually multicellular
(D) In root the multicellular epidermal hairs are called trichomes.
Which of the following statements are correct ?
(1) A and B (2) B and C (3) A and C (4) B, C and D

9. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) :
Assertion A : Both apical meristems and intercalary meristems are primary meristems
Reason R : They appear early in life of a plant and contribute to the formation of primary plant body.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explaination of A.
(3) A is true but R is false
(4) A is false and R is true

10. Xylem element with thickened walls and obliterated central lumen is :
(1) Xylem fibre (2) Vessels
(3) Tracheids (4) Xylem parenchyma

11. The vascular bundles in a dorsiventral leaf are :-
(1) collateral open with xylem towards adaxial epidermis and phloem towards abaxial epidermis.
(2) collateral closed with xylem towards adaxial epidermis and phloem towards abaxial epidermis.
(3) collateral closed with xylem towards abaxial epidermis and phloem towards adaxial epidermis
(4) radial with protoxylem towards periphery and metaxylem towards the centre

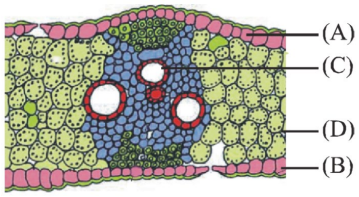
12. Which one of the following statements is correct, for "Bundle sheath of monocot leaves is similar to that of monocot stem", as both of them :-
(1) Possess outer layer of chlorenchyma and inner layer of thick-walled cells without chloroplasts
(2) Possess extensions made up of sclerenchyma
(3) Resemble the endodermis in possession of casparian strips
(4) Encircle the vascular bundles, which are conjoint and collateral

13. In a dicot leaf :-
(1) Stomata are found usually in equal number on both the surfaces
(2) Leaf mesophyll is undifferentiated
(3) Palisade parenchyma is placed adaxially
(4) Spongy parenchyma is placed adaxially

14. In temperate regions, the climatic conditions are not uniform through the year, in the spring season,cambium is very active and produces a large number of xylem elements having vessels with wider cavities.The wood formed during this season is called :-
(1) Late wood (2) Heart wood (3) Sap wood (4) Early wood

15. The____is the centermost tissue in a dicot stem :-
(1) Xylem (2) Pith (3) Pericycle (4) Conjuctive tissue

16. Given below is the diagrammatic view of a monocot leaf. Identify the parts labelled A, B, Cand D and select the right option :-



	Part-A	**Part-B**	**Part-C**	**Part-D**
(1)	Abaxial epidermis	Adaxial epidermis	Xylem	leaf mesophyll
(2)	Adaxial epidermis	Abaxial epidermis	Xylem	Leaf mesophyll
(3)	Adaxial epidermis	Abaxial epidermis	Phloem	Spongy mesophyll
(4)	Abaxial epidermis	Adaxial epidermis	Phloem	Pallisade mesophyll

17. Read the following statements and find out the **correct** set of statements :

A. Companion cells help in maintaining the pressure gradient in the sieve tubes.

B. Gymnosperm lack vessels in their xylem.

C. The xylem vessels are devoid of nucleus.

D. Xylem fibres may be septate or aseptate.

E. A mature sieve element in phloem possesses cytoplasm, vacuole and nucleus.

Choose the **correct** answer from the options given below:

(1) A, B, D, E only (2) A, B, C, D only

(3) C, D, E only (4) B, C, D, E only

18. Two statements are given below.

Statement-I : Stomatal apparatus is the part of ground tissue system.

Statement-II : In the vascular bundle of monocot stem cambium is present between xylem and phloem.

In the light of above statements choose the correct answer from the options given below.

(1) Both the statement-I and statement-II are correct

(2) Both the statement-I and statement-II are incorrect

(3) statement-I is correct but statement-II is incorrect

(4) statement-I is incorrect but statement-II is correct

19. Instead of sieve tube, in phloem of gymnosperm __A__ are present :

(1) Sieve cells (2) Companionm cells

(3) Albuminous cells (4) Fibres

20. The radial conduction of water takes place by -

(1) Tracheids (2) Vessels

(3) Ray Parenchymata (4) Sieve tubes

ANSWER KEY															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**2**	**3**	**2**	**1**	**2**	**3**	**3**	**1**	**1**	**2**	**4**	**3**	**4**	**2**
Que.	16	17	18	19	20										
Ans.	**2**	**2**	**2**	**1**	**3**										

STRUCTURAL ORGANISATION OF ANIMALS (ANIMAL TISSUES)

1. Choose the correct option with respect to the type of tissue and their appearance.

	Tissues	Appearance
(1)	Cuboidal epithelium	Flat with regular boundaries
(2)	Squamous epithelium	Flat with irregular boundaries
(3)	Columnar epithelium	Single layered, cube like cells
(4)	Cuboidal epithelium	Multi layered, cuboidal and slender cells

2. Compound epithelium

I. plays limited role in secretion and absorption. II. present in tubular part of nephron.

III. covers only the moist surface of skin. IV. are found in linings of pancreatic duct.

Choose the correct option from the following.

(1) Statements I and II are true, rest are false (2) Statements III and IV are true, rest are false

(3) Statements I and IV are true, rest are false (4) Only statement I is true, rest all other statement are false

3. Read the following statements carefully and select the option that correctly fills the blanks.

M. Specialised connective tissues includes cartilage, ...A....., bone and blood.

N. Cells secrete fibres of structural protein called .....B.....

O. The fibre provides .....C..... and strength to the tissues.

P. The connective tissue cells also secrete .....D..... which accumulate between cell and fibres and act as matrix.

	A	B	C	D
(1)	areolar	elastic	support	monosaccharide
(2)	fibre	enzyme	elasticity	disaccharide
(3)	collagen	fibroblast	strength	disaccharide
(4)	adipose	elastin	flexibility	polysaccharide

4. Choose the correct option regarding adipose tissue.

I. It is a modified loose connective tissue composed of adipocytes.

II. Its main role is to insulate the body and also provide cushion around visceral organs.

III. It is a loose connective tissues whose fibres are arranged in semifluid ground substance.

(1) I, II, III are correct (2) Only I and II are correct

(3) Only I and III are correct (4) Only II and III are correct

5. Which of the following statements in incorrect regarding connective tissues?

(1) The cells of cartilage tissue are specialised to store fats

(2) In the dense regular connective tissue, the collagen fibres are present in rows

(3) The areolar tissue serves as a support framework for epithelium

(4) They secrete the polysaccharide which accumulate between cells and fibres.

6. Which of the following statement is incorrect?

(1) All major functions like digestion and respiration are performed by a single cell in unicellular organisms

(2) The human body is composed of billions of cells to perform various regulatory functions

(3) The cells form an organ and helps in regulatory function in a random pattern

(4) Organ system includes respiratory system, digestive system, etc. which perform crucial functions for our survival as a whole.

7. The tissues present in the inner surface of Fallopian tube

A. are columnar. B. are branched.

C. Helps in embryo movement. D. are found in hollow organs and bear cilia.

E. have nuclei present at the base. F. are fusiform

Choose the correct option from the following.

(1) A, D, E, F (2) A, C, D, E (3) A, B, D, F (4) A, C, E, F

8. Pick the odd one in each series with respect to cell junction and select the correct option.

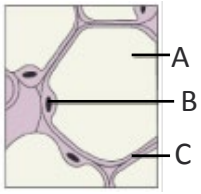
(i) Prevent Leakage, Fusion of cell membrane, Tight junction, Ducts

(ii) Tight junction, Cementing, Neighbouring cell, Gland cluster

(iii) Cell adjoining, Gap junction, Ion transfer, Buccal cavity

	(i)	(ii)	(iii)
(1)	Leakage	Tight junction	Ion transfer
(2)	Ducts	Gland cluster	Buccal cavity
(3)	Leakage	Gland cluster	Gap junction
(4)	Tight junction	Neighbouring cell	Buccal cavity

9. Carefully analyse A, B and C in the given diagram.



Which of the following option is incorrect w.r.t. the given diagram?

(1) Loosely arranged cells and fibres in a semi-fluid substance.

(2) B represent the nucleus which is the most crucial part for the cellular function.

(3) A represents the vacuole space of the cell containing fluid and metabolic compounds.

(4) The given cell is found beneath the skin.

10. Select the incorrect match.

(1)	Fibroblast	Matrix and fibres
(2)	Mast cell	Histamine, heparin, serotonin
(3)	Adipocytes	Leptin hormone
(4)	Osteocyte	Located in matrix of cartilage

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**2**	**3**	**4**	**1**	**1**	**3**	**2**	**2**	**3**	**4**

STRUCTURAL ORGANISATION OF ANIMALS (COCKROACH AND FROG)

1. Which of the following is the correct sequence representing the parts of alimentary canal of cockroach starting from mouth?

(1) Pharynx → Oesophagus → Gizzard → Ileum → Crop → Colon → Rectum

(2) Pharynx → Oesophagus → Crop → Gizzard → Ileum → Colon → Rectum

(3) Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum

(4) Pharynx → Oesophagus → Gizzard → Crop → Ileum → Colon → Rectum

2. Read the following statements regarding cockroach.

I. The rectum opens out through anus.

II. In male cockroach, a characteristic mushroom shaped gland is present in the 6th-7th abdominal segments which functions as an accessory reproductive gland.

III. Cockroach is ureotelic.

IV. Blood from sinuses enters heart through ostia and is pumped anteriorly to Aorta.

Choose the **incorrect** statement from the options given below.

(1) Statements I and II are correct

(2) Statement III is incorrect

(3) Statement III is correct

(4) Statements III and IV are incorrect

3. Which of the following statement is correct regarding cockroach?

(1) Heat is oval in shape

(2) Each eye consists of about 10000 hexagonal ommatidia

(3) The 10th segment in both the sexes bear a pair of jointed filaments structure called anal cerci

(4) Heart is differentiated into funnel shaped chambers with setae on either sides.

4. Consider the following statements and select the correct option stating which ones are true (T) and which ones are false (F).

I. In female cockroach, every ovary is formed of a group of ten ovarian tubules or ovarioles, containing a chain of developing ova.

II. In cockroach, the nymph grows by moulting about 13 times to reach the adult form.

III. In male cockroach, genital pouch or chamber lies at the hind end of abdomen bounded dorsally by 9th and 10th terga and ventrally by the 9th sternum.

IV. In cockroach the haemolymph is composed of colourless plasma and haemocytes.

	I	II	III	IV
(1)	T	F	T	F
(2)	F	T	T	T
(3)	T	T	F	F
(4)	F	T	F	F

5. The anal styles in cockroaches are present on -

(1) 10th tergite of female

(2) 9th sternite of male

(3) 9th sternite of female

(4) 10th tergite of male

6. "Frogs exhibits sexual dimorphism."
Which of the following options justifies the given statement?
(1) Female frogs attracts males by their specialised vocal sacs.
(2) Male frogs have sound producing vocal sacs
(3) Female frog have webbed feets
(4) Both female and male frogs exhibits five digit at the end of hindlimbs.

7. Read the following statements (P-S) and select the options that correctly fills the blanks.
P. The alimentary canal of frog is short because frog are .....(i).....
Q. .....(ii)..... store the excess bile juice, secreted by the liver.
R. Digestion takes place by the action of .....(iii)..... and .....(iv)..... secreted from the walls of stomach.
S. Finger-like folds in the inner walls of intestine called .....(v)..... and .....(vi)..... help in absorption of digested food.

	(i)	(ii)	(iii)	(iv)	(v)	(vi)
(1)	heterotrophs	Intestine	amylase	tryptophan	villi	microbilli
(2)	herbivores	Pancreas	pepsin	HCl	cloaca	microvilli
(3)	omnivores	Rectum	HCl	renin	intestine	cilia
(4)	carnivores	Gall bladder	HCl	gastric juice	villi	microvilli

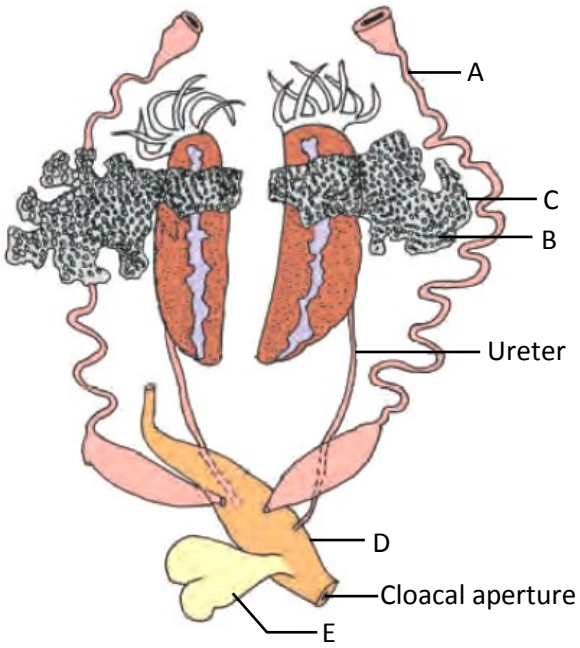
8. The correct sequence of the part of the alimentary canal of frog starting from mouth is
(1) Mouth → Intestine → Stomach → Bladder's canal → Rectum → Anus
(2) Pharynx → Oesophagus → Stomach → Duodenum → Rectum → Cloaca
(3) Mouth → Larynx → Pharynx → Stomach → Intestine → Rectum
(4) Mouth → Oesophagus → Stomach → Intestine → Rectum → Anus → Cloaca

9. Read the following statements carefully.
A. During aestivation and hibernating in frogs, gaseous exchange takes place through .....(i).....
B. The lungs are elongated, sac like structure present in .....(ii).....
C. The vascular system of frog is .....(iii).....
D. The lymphatic system consists of lymph, ...(iv)..... and .....(v).....

	(i)	**(ii)**	**(iii)**	**(iv)**	**(v)**
(1)	cuticle	lower thorax	open type	nerve channels	blood
(2)	skin	upper thorax	closed type	lymph channel	lymph nodes
(3)	endodermis	thymus	open type	vessels	bone
(4)	skin	opposite vertebral column	closed type	nodes	vessels

10. Read the statements regarding frog, which of the following statements is/are incorrect and correct?
I. 10 pairs of cranial nerves arise from the brain.
II. Midbrain is characterised by the paired cerebral hemispheres.
III. A pair of ovaries are situated near the kidney having functional connection with it.
IV. The medulla oblongata passes out through the foramen magnum and continues into spinal cord.
(1) Statements I and III are correct, while statements II and IV are incorrect.
(2) Statements I and II are incorrect, while statements III and IV are correct
(3) Statements I and IV are correct, while statements II and III are incorrect
(4) Statements I and III are incorrect, while statements II and IV are correct

11. Observe the following figure and identify A to E.



How many of the following are true regarding the figure given above?

P. B is situated near the kidneys, with no functional connect with kidneys.

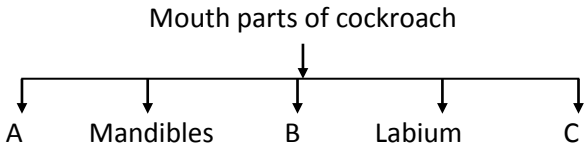
Q. A arises from the ovary and opens into the cloaca separately.

R. Around 25000-30000 of C are produced at a time after maturity.

S. D is the common chamber for urinary tract and alimentary canal.

(1) One (2) Two (3) Three (4) Four

12. Examine the given flowchart and identify A, B and C.



	A	B	C
(1)	Ocellus	Typhlosole	Maxillae
(2)	Labrum	Maxillae	Hypopharynx
(3)	Labullus	Peristomium	Prostopharynx
(4)	Labium	Maxillae	Hypopharynx

13. Identify the incorrect statement with respect to *Periplaneta americana.*

I. In vascular system of cockroaches, the blood vessels are poorly developed and open into the space called haemocoel.

II. There are 16 very long Malpighian tubules present at the junctions of midgut and hindgut.

III. Heart of cockroach consist of elongated muscular tube lying along mid dorsal line of thorax and abdomen.

IV. Frog maintains ecological balance although they are also eaten by man in some countires.

(1) I and III (2) I and II (3) III and IV (4) II and IV

14. Read the given statements w.r.t. X.

I. It is lined by glandular and ciliated cells.

II. It absorbs nitrogenous waste products from haemocoel and convert them into uric acid, which is excreted out through the hindgut.

Which of the following structures of cockroach does X refer to ?

(1) Malpighian tubule (2) Hepatic caecae

(3) Trachea (4) Tergum

15. Consider the following statements and select the option which correctly fills up the blanks.

P. In cockroach, the average number of ootheca formed are .....

Q. The development of *Periplaneta americana* is .....

R. The nymph of *Periplaneta americana* undergo ..... about 13 times before becoming an adult.

	P	Q	R
(1)	14-16	homolatabolous	metamorphosis
(2)	9-10	paurometabolous	moulting
(3)	5-6	ametabolous	morting
(4)	13-15	heamimetabolous	mosting

16. Cockroach has no RBCs or haemoglobin because -

(1) Gases are transported and dissolved in plasma.

(2) It respires anaerobically

(3) Its blood does not transport gases since it has tracheal respiration

(4) It does not respire

17. Read the following characteristics of the frog.

I. Smooth skin.

II. Bulged eyes and four digits or limbs.

III. Tympanic membrane.

IV. Body division includes head, neck and trunk.

V. Pale yellow pigment on dorsal side.

Choose the correct option from the following.

(1) Features in statements I, III and IV are true, rest are false

(2) Features in statements I, II and III are true, rest are false

(3) Features in statements III and IV are true, rest are false

(4) Only the feature in statement II is true, rest are false

18. The male and female frog can be distinguished by the presence of .....(i)..... and .....(ii).....

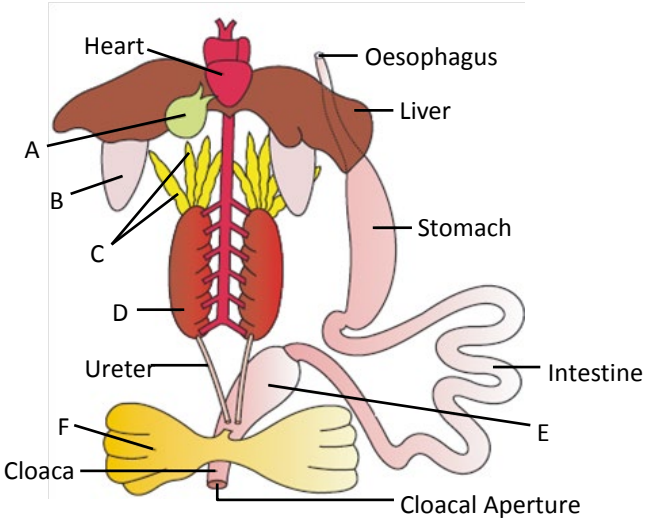
(1) (i) vocal sacs, (ii) copulatory pad on second digit of forelimbs.

(2) (i) vocal sacs, (ii) copulatory pad on first digit of forelimbs.

(3) (i) Nictitating membrance, (ii) copulatory pad on first forelimbs

(4) (i) vocal sac, (ii) nictitating membrane

19. Given below is the diagram of internal organs of frog. Identify A to F.



Which of the following matched pair is correct w.r.t. the given diagram and its features?

(1) A-Store house of bile

(2) B-Produce pancreatic juice

(3) D-Uriniferous tubules

(4) F-Storage of excretory liquid

20. Which of the following statements is/are incorrect?

I. Frog respire through skin under water which is called cutaneous respiration.

II. In aquatic respiration, air enters into buccal cavity and then travels towards gills.

III. During hibernation, the frog exhibits gaseous exchange by cutaneous respiration.

(1) Statement II and III are incorrect

(2) Statement II is incorrect

(3) Statements I and II are incorrect

(4) Statements I and II are incorrect

21. Consider the following four statements I-IV and select the correct option stating which one is true (T) and which one is false (F) w.r.t. the vascular system of frog.

I. Special venous connection between liver and intestine as well as kidney and lower parts of the body are present in frogs.

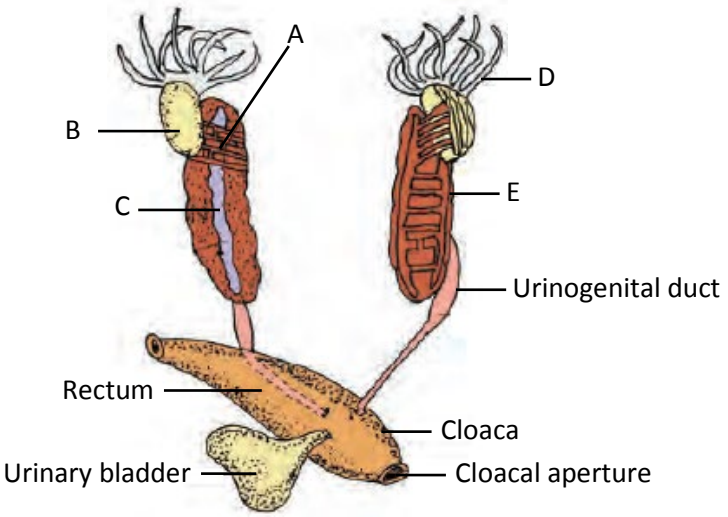
II. A triangular structure called conus arteriosus joins the right atrium.

III. The veins collect blood from different parts of body to the heart and form the venous system.

IV. The blood carry nutrients, gases and water to the respective sites during circulation.

	I	II	III	IV
(1)	F	F	T	T
(2)	F	T	T	F
(3)	T	T	F	F
(4)	T	F	T	T

22. Identify A, B, C, D and E in the given figure of reproductive system of frog.



Which of the following option is correctly matched according to the given diagram and their features?

	Name	**Feature (I)**	**Feature (II)**
(1)	A-Bidder's canal	Ventral side of testes	Secretory function
(2)	B-Testis (Ovoids)	Adhere to upper part of kidneys	Mesorchium
(3)	C-Adrenal gland	No functional connection	Metamorphosis
(4)	E-Kidney	Double fold of peritoneum	Balancing equilibrium

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**2**	**3**	**2**	**2**	**2**	**4**	**2**	**2**	**3**	**2**	**2**	**4**	**1**	**2**
Que.	16	17	18	19	20	21	22								
Ans.	**3**	**2**	**2**	**1**	**2**	**4**	**2**								

CELL : THE UNIT OF LIFE

1. Different cells have different sizes. Arrange the following in an ascending order of their size. Choose the correct option among the following:-

(i) Mycoplasma (ii) Ostrich eggs

(iii) Human RBC (iv) Bacteria

(1) i, ii, iii, iv (2) i, iv, iii, ii

(3) ii, i, iii, iv (4) iii, ii, i, iv

2. **Assertion :-** The cytoplasm is the main arena of cellular activities in cell.

Reason :- The various chemical reactions occurs in cytoplasm to keep the cell in the living state.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

3. How many structures of cell in list given below are single membrane bound?

Centriole, Mitochondria, Ribosome, Plastid, Lysosome, ER, Golgi apparatus, Nucleus

(1) Three (2) Four

(3) Two (4) One

4. A researcher made an interesting observations about a protein made by the rough ER and eventually used to build a cell's plasma membrane. The protein in the membrane was actually slightly different from the protein made in the ER. The protein was probably changed in the :-

(1) Golgi apparatus (2) Smooth ER

(3) Mitochondria (4) Nucleus

5. Select the **correct** statements.

A. Cell wall helps in cell-to-cell interaction and provides barrier to undesirable macromolecules.

B. The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together.

C. SER is frequently observed in the cells actively involved in protein synthesis and secretion.

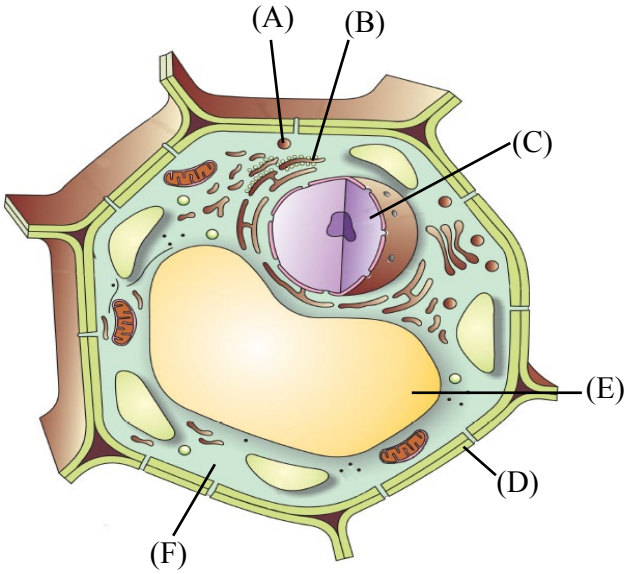
D. The cis and the trans faces of the organelle are entirely different, and are not interconnected.

Choose the correct answer from the option given bellow:

(1) A and B only (2) A and D only

(3) A, B and C only (4) A, C and D only

6. Find the correct labelling for given plant cell ?



	A	B	C	D	E	F
(1)	Lysosome	SER	Nucleus	Cell wall	Vacuole	Cytoplasm
(2)	Lysosome	RER	Nucleus	Cell wall	Vacuole	Cytoplasm
(3)	Lysosome	RER	Vacuole	Cell wall	Nucleus	Cytoplasm
(4)	Lysosome	SER	Nucleus	Cell wall	Vacuole	Peroxisome

7. **Assertion (A):** Chloroplast and mitochondria are semi-autonomous cell organelles.

Reason (R): Both of these organelles have their own genetic material and protein synthesis machinery & can synthesize their few proteins.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

8. **Assertion:** Axonemal microtubules are arranged in 9+2 array in cilia and flagella.

Reason: Axoneme usually has nine pair of triplets of radially arranged microtubules and a pair of centrally located microtubules.

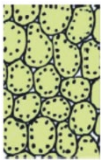
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

9. Match the columns I, II and III and choose the correct combination from the options given :-

Column-I		Column-II		Column-III	
a.		1.	Mesophyll cell	K.	Elongated
b.		2.	Tracheid	L.	Round and biconcave
c.		3.	Red blood cells	M.	Amoeboid
d.		4.	White blood cells	N.	Round and oval

(1) a-3-M, b-4-L, c-1-K, d-2-N (2) a-4-M, b-3-L, c-2-K, d-1-N

(3) a-3-L, b-4-M, c-1-N, d-2-K (4) a-4-L, b-3-M, c-2-N, d-1-K

10. Given below are two statements :

Statement-I : In eukaryotic cells, there is an extensive compartmentalisation of cytoplasm through the presence of membrane bound organelles.

Statement-II : The quasi-fluid nature of proteins enables lateral movement of lipid within the overall bilayer.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement-I and Statement-II are false. (2) Statement-I is correct but Statement-II is false.

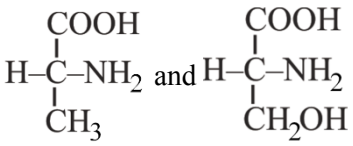
(3) Statement-I incorrect but Statement-II is true. (4) Both Statement-I and Statement-II are true.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**2**	**1**	**1**	**1**	**1**	**2**	**4**	**3**	**3**	**2**

BIOMOLECULES

1. Given below are the structure of two different amino acids

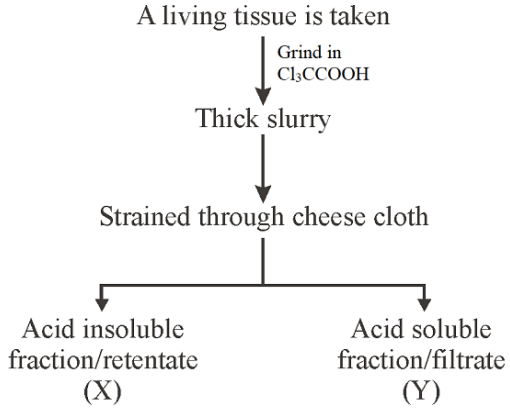


These amino acids show which type of nature respectively ?

(1) Acidic, Alkaline (2) Alkaline, Neutral

(3) Neutral, Neutral (4) Alkaline, Alkaline

2. Study the flow chart given below and identify X and Y according to the statements (i-v) provided :-



(i) Monomers are present

(ii) Polymers are abundant

(iii) Molecular weight in the range of 10 thousand daltons.

(iv) Molecular weight between 18-800 Daltons

(v) Amino acids, nucleotides monosaccharides

	X	Y
(1)	(ii), (iii)	(i), (iv), (v)
(2)	(i), (ii)	(iii), (iv), (v)
(3)	(iii), (iv)	(i), (ii), (v)
(4)	(i), (iv), (v)	(ii), (iii)

3. The correct order of chemical composition of living tissues/cells in term of percentage of the total cellular mass is :-

(1) Nucleic acid > Proteins > H_2O > Carbohydrates > lons > Lipids

(2) H_2O > Proteins > Nucleic acid > Carbohydrates > Lipids > lons

(3) H_2O > Proteins > Carbohydrates > Nucleic acid > Lipids > lons

(4) Lipids > lons > Carbohydrates > H_2O > Proteins > Nucleic acid

4. **Assertion :** Amino acids are substituted methanes.

Reason : Amino acid contain an amino group and an acidic group.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

5. A list of different metabolites is given below. In this list find out the number of primary & secondary metabolites, respectively & choose the correct option:
Alkaloids, Abrin, Ricin, Carotenoids, Amino acids, Glucose, Fructose, Fatty acids, Thymine, Uracil, Lectins, Drugs.
(1) Five and Seven (2) Six and Six (3) Seven and five (4) Four and Eight

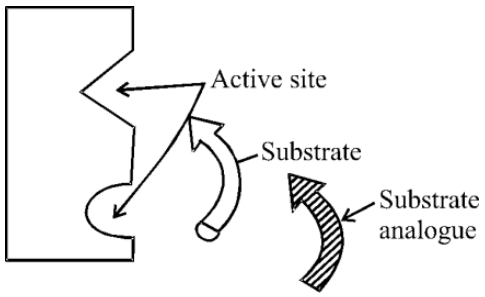
6. Find out the **correct** match :-

	I	II	III
	Class	Enzyme	Function
(1)	II	Lyases	$S-G+S' \rightarrow S+S'-G$
(2)	IV	Isomerases	$\begin{matrix} X & Y \\ \vert & \vert \\ C-C \end{matrix} \rightarrow X-Y+C=C$
(3)	I	Transferases	$S-G+S' \rightarrow S+S'-G$
(4)	III	Hydrolases	Hydrolysis of ester bond

7. Which of the following statements regarding cofactors is **not** correct ?
(1) Every prosthetic group is a cofactor but every cofactor is not a prosthetic group
(2) Catalytic activity is lost when the cofactor is removed from the enzyme
(3) Every coenzyme is a cofactor and every cofactor is a coenzyme
(4) In the catalase enzyme, haem group work as a cofactor

8. One molecule of enzyme converts the 5 molecules of substrate into product in 5 minutes. 30 molecules of substrate and two molecules of enzyme are mixed in a test tube. After 10 minutes the test tube will have :-
(1) Products and 10 uncatalysed molecules of substrate
(2) Only products
(3) 10 uncatalysed molecules of substrate, enzymes and products
(4) Products and enzymes

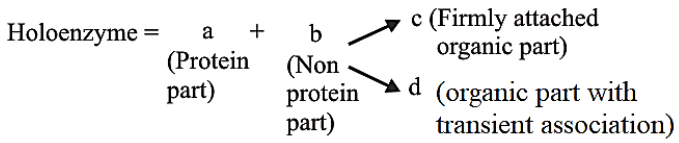
9.



Above diagram is the indicative diagram of a type of inhibition. Choose the correct match :-

	Type of Inhibition	Substrate analogue	Enzyme
(1)	Non-Competitive reversible	Malonate	Succinate
(2)	Competitive	Malonate	Succinic dehydrogenase
(3)	Non- Competitive irreversible	Succinate dehydrogenase	Succinate
(4)	Competitive	Succinate	Succinic dehydrogenase

10.



Choose the correct terms for a to d from the following :-

(1)	Cofactor	Activator	Apoenzyme	Prosthetic Group
(2)	Activator	Cofactor	Apoenzyme	Prosthetic Group
(3)	Apoenzyme	Coenzyme	Prosthetic group	Cofactor
(4)	Apoenzyme	Cofactor	Prosthetic group	Coenzyme

ANSWER KEY										
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**3**	**1**	**2**	**2**	**2**	**4**	**3**	**3**	**2**	**4**

CELL CYCLE & CELL DIVISION

1. Match the **List-I** with **List-II**.

List I		**List II**	
A	S-phase	1	Proteins are synthesised
B	G_2 -phase	2	Inactive phase
C	Quiescent stage	3	Interval between mitosis and initiation of DNA replication
D	G1 -phase	4	DNA replication

Choose the correct answer from the options given below.

(1) A-3, B-2, C-1, D-4 (2) A-4, B-2, C-3, D-1

(3) A-4, B-1, C-2, D-3 (4) A-2, B-4, C-3, D-1

2. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The individual chromosomes can no longer be seen and chromatin material tend to collect in a mass at two poles in telophase.

Reason (R) : The chromosomes that have reached their respective poles decondense and lose their individuality.

In the light of the above statements. Choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct.

(3) (A) and (R) both are incorrect.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

3. If a cell has 10 Pg DNA in G_1-stage of interphase, what will be the amount of DNA in each daughter cell after meiosis-I :-

(1) 10 Pg (2) 20 Pg

(3) 40 Pg (4) 5 Pg

4. Read the following statements :-

Statement-I : During cell division, DNA synthesis occurs only during one specific stage in the cell cycle, known as G_2 phase.

Statement-II : During cell division, centriole replicates in S-phase of cell cycle in plant cells.

Choose the correct option :–

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

5. Read the following statements :-

(i) Prophase is marked by the initiation of condensation of chromatin material.

(ii) The chromosomal material becomes untangled during the process of chromatin condensation.

(iii) In the S and G_2 phase the new DNA molecules formed are not distinct but interwined.

(iv) Nuclear envelope remains intact throughout the prophase

(v) At the end of prophase, when viewed under microscope, a cell shows distinct Golgi body, ER and nucleolus

How many of the above statements are **correct**?

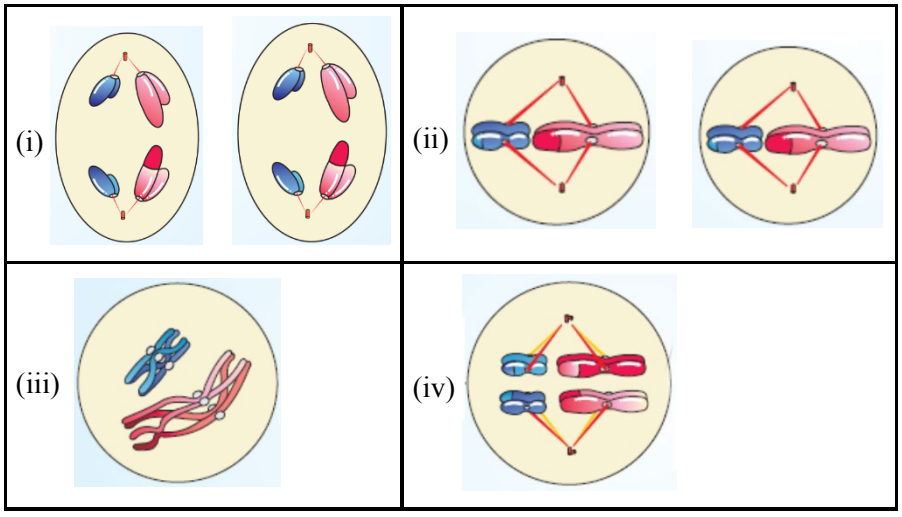
(1) Three (2) Four (3) Two (4) One

6. If the initial amount of DNA in a cell is represented as 2C then the amount of DNA in this cell immediately after S phase, at metaphase and in anaphase, respectively, should be :-

(1) 4C, 4C and 2C (2) 4C, 4C and 4C

(3) 2C, 4C and 4C (4) 2C, 4C and 2C

7. Arrange the different stages (i), (ii), (iii) and (iv) of meiosis in sequence :



(1) (i) → (iv) → (ii) → (iii) (2) (iii) → (iv) → (ii) → (i)

(3) (iv) → (iii) → (ii) → (i) (4) (ii) → (i) → (iii) → (iv)

8. **Assertion :-** Crossing over is the exchange of genetic material between two homologous chromosomes.

Reason :- Crossing over is an enzyme mediated process and enzyme involved is recombinase.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

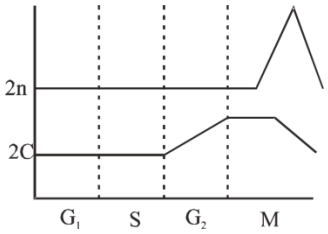
9. If chromosome number and DNA content of unfertilised egg cell is 20 and 100 Pg respectively, then how many chromosomes and how much DNA will present during G_2 phase and after meiosis-I respectively ?

(1) 40, 400, 20, 200 (2) 20, 200, 40, 400

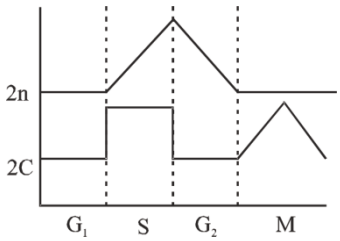
(3) 40, 200, 40, 200 (4) 20, 400, 20, 200

10. Which of the following graph correctly represents DNA content (2C) and chromosome number (2n) of a diploid cell undergoing different stages of cell cycle ?

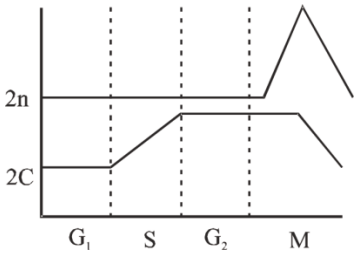
(1)



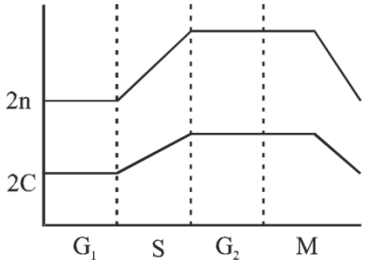
(2)



(3)



(4)



ANSWER KEY											
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	**3**	**4**	**1**	**4**	**1**	**2**	**2**	**2**	**1**	**3**	

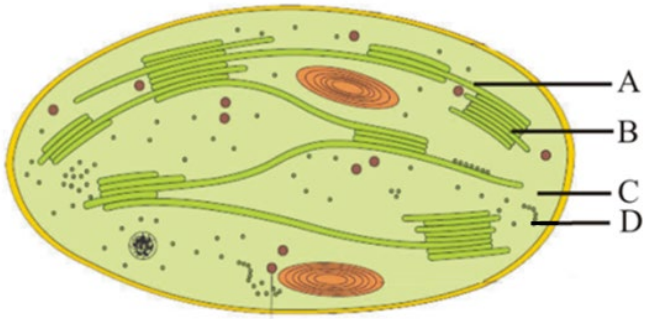
PHOTOSYNTHESIS IN HIGHER PLANTS

1. Match the columns :-

Column-I		Column-II	
A.	Joseph Priestley	(i)	Green plant parts synthesise sugar
B.	Jan Ingenhousz	(ii)	In photosynthesis O_2 comes from H_2O
C.	Julius von Sachs	(iii)	Discovered oxygen
D.	Cornelius Van Niel	(iv)	Sunlight is essential to the plant process that purifies fouled air

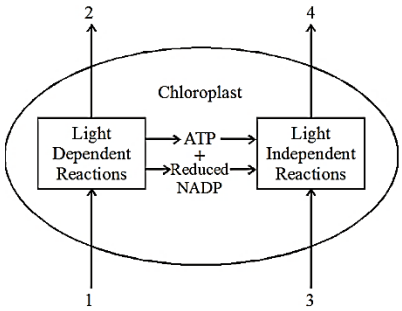
(1) A = (i), B = (iv), C = (iii), D = (ii)
(2) A = (iv), B = (ii), C = (iii), D = (i)
(3) A = (ii), B = (i), C = (iii), D = (iv)
(4) A = (iii), B = (iv), C = (i), D = (ii)

2. Which one is **incorrect** option for given diagram :-



(1) **A**–Only cyclic photo photophosphorylation takes place.
(2) **B**–Site for the synthesis of ATP only
(3) **C**–Site of carbon reactions
(4) **D**–Site of protein synthesis

3. The diagram shows the movement of substances in and out of a chloroplast :



What do labels 1 to 4 represent ?

	1	2	3	4
(1)	CO_2	ATP	H_2O	Starch
(2)	CO_2	H_2O	Sugar	O_2
(3)	H_2O	O_2	CO_2	Sugar
(4)	Sugar	H_2O	ATP	O_2

4. Accessory pigments of photosynthesis :

(1) Are chlorophyll-b, xanthophylls and carotenes

(2) Absorb light and transfer electron to chlorophyll-a

(3) Enable only a narrow range of wavelength of incoming light to be utilised for photosynthesis and also protects chlorophyll-a from photo-oxidation

(4) All are correct

5. Select the **correct** statements from the following :-

(A) Accessory pigments enable a wider range of wavelength of incoming light to be utilised for photosynthesis.

(B) The membrane system consisting of grana and stroma lamella is responsible for trapping light energy.

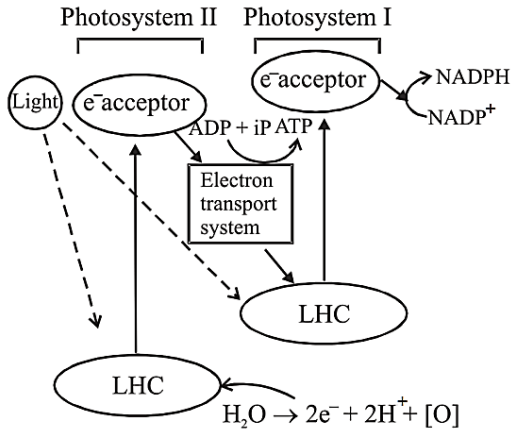
(C) In grana, enzymatic reactions incorporate CO_2, into the plant leading to the synthesis of sugar.

(D) Chloroplast align themselves along the walls of the mesophyll cells, to get optimum quantity of incident light.

(1) A, B only (2) B, D only

(3) B, C, D only (4) A, B, D only

6.



Which of the following is the location of process shown in above diagram ?

(1) Lamellae which form stacks in the chloroplast

(2) The fluid part of the chloroplast

(3) Lamellae which connect the grana lamellae

(4) Space between outer and inner membrane of chloroplast.

7. The Z scheme of photophosphorylation follows the following sequence :-

$$\text{PS II} \xrightarrow[A]{} e^- \text{ acceptor} \xrightarrow[B]{} \text{ETS} \xrightarrow[B]{} \text{PS I} \xrightarrow[C]{} e^- \text{ acceptor} \xrightarrow[D]{} NADP^+$$

Which of the following options is correct for A, B, C, D transfer of electrons ?

	A	B	C	D
(1)	Uphill	Downhill	Uphill	Downhill
(2)	Downhill	Uphill	Downhill	Uphill
(3)	Downhill	Uphill	Uphill	Downhill
(4)	Uphill	Downhill	Downhill	Uphill

8. Proton gradient required for ATP generation in cyclic electron transport chain arise due to:

(1) PQ mediated shifting of H^+ from stroma to lumen.

(2) Photolysis of water

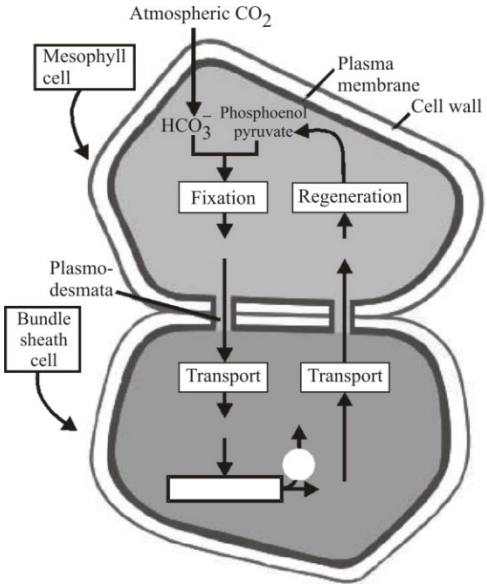
(3) NADP reductase mediated consumption of H^+ in stroma.

(4) All the above

9. Which of the following amount of PGAL molecules is not sufficient to regenerate 10 RuBP ?

(1) 20 (2) 25 (3) 15 (4) 30

10. Observe the diagram given below :



Which statement is **correct** for the above pathway ?

(1) Photosynthetic decarboxylation takes place in mesophyll cells.

(2) C_3 acid is transported from mesophyll cells to bundle sheath cells.

(3) Fixation of CO_2 catalysed by RuBisCO enzyme occurs in bundle sheath cell.

(4) Secondary CO_2 fixation occurs in mesophyll cells.

11. **Assertion:** Green plants carry out 'Photosynthesis' which is a 'Physico-chemical process'.

Reason: Green plants use light energy to drive the synthesis of organic compounds.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

12. **Statement-I :** In the half leaf experiment a part of leaf in enclosed in a test tube containing some KOH soaked cotton.

Statement-II : Half leaf experiment shows that light is required for photosynthesis.

(1) Statement-I & Statement-II both are correct.

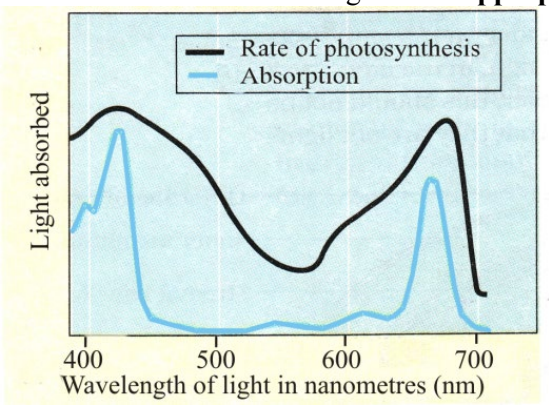
(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Statement-I & Statement-II both are incorrect.

13. **Assertion:** Chlorophyll 'a' is the chief pigment associated with photosynthesis.
Reason: The wavelength at which there is maximum absorption by chlorophyll 'a' i.e. in the blue and the red region also show higher rate of photosynthesis.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

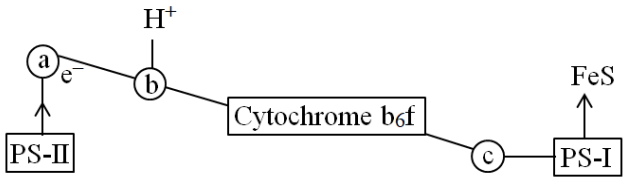
14. Represented below is the graph showing action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll 'a'. Which one of the following is **most appropriate** for this?



(1) There is a complete one to one overlap between these absorption and action spectrum
(2) By this overlapping it can be concluded that chlorophyll 'a' is the chief pigment of photosynthesis
(3) This overlapping shows that only chlorophyll 'a' absorb the entire light energy
(4) These graphs show that photosynthesis occurs only in the wavelengths of blue and red light

15. Paper chromatography is used to separate the leaf pigments. On a chromatogram yellowish green colour represent the :-
(1) Chlorophyll-a
(2) Chlorophyll-b
(3) Carotenoids
(4) Xanthophylls

16.



In above diagram **(b)** and **(c)** are :-
(1) **b**-Plastoquinone, **c**-FRS protein
(2) **b**-Plastoquinone, **c**-Plastocyanin
(3) **b**-Plastocyanin, **c**-Plastoquinone
(4) **b**-Plastoquinone, **c**-Pheophytin

17. Consider the following four statements A, B, C and D, and select the option having **correct** statements :-
(A) Z-scheme of light reaction takes place in presence of PS I only.
(B) Only PS I is functional in cyclic photophosphorylation.
(C) Cyclic Photophosphorylation results into synthesis of ATP and NADPH.
(D) Stroma lamellae membranes lack PS II as well as NADP reductase enzyme.
Options :
(1) C and D
(2) B and D
(3) A and B
(4) B and C

18. How many H_2O molecules are split at water splitting complex for evolution of 6 molecules of oxygen ?

(1) 4 (2) 6 (3) 12 (4) 18

19. Choose the **correct** option with respect to location and raw material for biosynthetic pathway in photosynthesis in higher plants.

	Location	**Raw Material**
(1)	Stroma	NADH, ATP, CO_2,H_2O
(2)	Stroma lamella	ADP, NADPH, CO_2, H_2O
(3)	Grana	ATP, NADPH, CO_2, H_2O
(4)	Stroma	ATP, NADPH, CO_2, H_2O

20. Which of the following ratio is correct for the production of one molecule of sucrose through Calvin cycle?

	CO_2	**:**	**ATP**	**:**	**NADPH**
(1)	12	:	12	:	12
(2)	6	:	18	:	12
(3)	6	:	12	:	18
(4)	12	:	36	:	24

ANSWER KEY															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**2**	**3**	**1**	**4**	**1**	**1**	**1**	**3**	**3**	**1**	**2**	**1**	**2**	**2**
Que.	16	17	18	19	20										
Ans.	**2**	**2**	**3**	**4**	**4**										

RESPIRATION IN PLANTS

1. ATP formation is not observed in which of the following?

(1) Conversion of PEP to pyruvic acid (2) Oxidative phophorylation

(3) Substrate level phosphorylation (4) Conversion of fructose 1, 6-bisphophate to PGAL

2. Consider the given statement with respect to glycolysis.

I. It causes partial oxidation of glucose (one molecule) to form 2 molecules of pyruvic acid and 2 ATP as net gain.

II. The process takes place in all living cells.

III. Glycolysis uses 2 ATP at every step.

IV. The scheme was given by Gustav Embden, Otto Meyerhof and J Parnas.

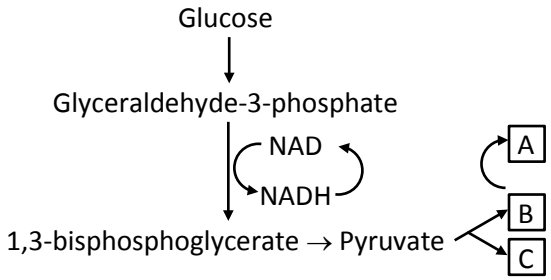
V. Pyruvic acid is the key product of glycolysis.

Choose the option containing correct statement(s).

(1) I, II and III (2) I, II, III and IV

(3) I, II, IV and V (4) Only I

3. Choose the correct combination of labelling the molecules involved in the pathway of anaerobic respiration in yeast.



(1) **A**-Ethanol, **B**-H_2O, **C**-Acetadegtde (2) **A**-CO_2, **B**-Ethanol, **C**-Acetaldehyde

(3) **A**-Acetaldehyde, **B**-CO_2, **C**-Ethanol (4) **A**-Ethanol, **B**-Acetaldehyde, **C**-CO_2

4. Which of the following are correctly matched?

A. Succinyl – Co-A → Succinate → SLP

B. α-ketoglutarate → Succinyl Co-A ⇒ Oxidative decarboxylation

C. Oxaloacetic acid → Citrate → Decarboxylation

D. Oxalosuccinate → α-ketoglutarate → Decarboxylation

(1) A, B and D (2) Only A and D (3) B and C (4) A, B, C and D

5. Consider the following statements with respect to Kreb's cycle.

I. Condensation of acetyl group with OAA and water to give citric acid is catalysed by citrate synthase.

II. 1 molecule of GTP that equals 2 ATP molecules, is synthesised during conversion of succinyl Co-A to succinic acid.

III. During the overall process of the TCA cycle, several intermediates are formed before the completion of the cycle.

IV. Continued oxidation of acetyl Co-A via Kreb's cycle requires continued replenishement of oxaloacetic acid a 4C compound.

V. NAD^+ is reduced to NADH + H^+ twice in the complete cycle.

Choose the **incorrect** set of statements.

(1) III and IV (2) I and II (3) II and V (4) IV and V

6. The e^- carrier molecules and cytochrome
(1) are reduced as they pass electrons on to next molecule.
(2) transfer electrons between the electron complexes
(3) shuttle protons to ATP synthase
(4) are found in outer mitochondrial membrane

7. The respiratory Quotient (RQ) of some of the compounds are 4, 1 and 0.7. These compounds are identified respectively as
(1) malic acid, palmitic acid and tripalmitin
(2) oxalic acid, carbohydrate and tripalmitin
(3) tripalmitin, malic acid and carbohydrate
(4) palmitic acid, carbohydrate and oxalic acid

8. **Statement–I :** Electrons from NADH produced in the mitochondrial cristae, during the TCA, are oxidised by NADH dehydrogenase, i.e., complex-II.

Statement–II : Cytochrome c is a small protein lying in the inner surface of the outer membrane that acts as a mobile carrier for e^- transfer between complex II and IV.

(1) Both statement I and statement II are correct
(2) Statement I is correct, but statement II is incorrect
(3) Statement I is incorrect, but statement II is correct
(4) Both statement I and statement II are incorrect

9. Match the following columns.

	Column-I (Compounds)		**Column-II (Examples)**
A.	4 C compound	1.	Acetyl Co-A
B.	2 C compound	2.	Pyruvate
C.	5 C compound	3.	Citric acid
D.	3 C compound	4.	a-ketoglutaric acid
		5.	Malic acid

Codes :

	A	B	C	D
(1)	2	5	3	1
(2)	3	1	4	2
(3)	5	1	4	2
(4)	5	3	1	2

10. Mitochondria are called powerhouse of the cell. Which of the following observations supports this statement?
(1) Mitochondria synthesise ATP
(2) Mitochondria have a double membrane
(3) The enzymes of the Krebs' cycle and the cytochromes are found in mitochondria matrix
(4) Mitochondria are found in almost all plants and animal cells

11. In order to enter the glycolytic pathway, sucrose is converted into glucose and fructose by the enzyme.
(1) invertase (2) zymase (3) isomerase (4) triose phosphatase

12. In which of the following reactions of glycolysis, oxidation takes place?
(1) Glucose-6-P to fructose-6-P
(2) Glyceraldehyde-3-phosphate to 1, 3-bisphosphoglycerate
(3) 1, 3-bisphosphoglycerate to 3-phophoglycerate
(4) 2-phosphoglycerate to phosphoglycerate

13. How many enzymes given below are responsible for conversion of pyruvic acid into CO_2 and ethanol.
Hexokinase, pyruvic acid decarboxylase, pyruvate kinase, alcohol dehydrogenase, lactate dehydrogenase.
(1) 2 (2) 3 (3) 4 (4) 5

14. Which one of the following reactions is an example of oxidative decarboxylation?
(1) Conversion of succinate to fumarate (2) Conversion of fumarate to malate
(3) Conversion of pyruvate to acetyl Co-A (4) Conversion of citrate to isocitrate

15. In which of the following, reduction of NAD^+ does not occur?
(1) Isocitric acid $\rightarrow$ α-ketoglutaric acid (2) Malic acid $\rightarrow$ Oxaloacetic acid
(3) Pyruvic acid $\rightarrow$ Acetyl Co-A (4) Succinic acid $\rightarrow$ Fumaric acid

16. Consider the following points.
I. Consumption of 2 ATP molecules.
II. Yield $2FADH_2$ molecules and 6 NADH molecules.
III. Linear pathway
IV. Enzymes participating, get dissolved in matrix.
How many of the above points are common to glycolysis and TCA?
(1) 1 (2) 3 (3) 2 (4) 0

17. The respiratory quotient during cellular respiration would depend on the
(1) nature of enzymes involved (2) nature of the substrate
(3) amount of carbon dioxide released (4) amount of oxygen utilised

18. **Assertion (A) :** Two turns of Kreb's cycle occur per glucose molecule used.
Reason (R) : 3 NADH, 1 $FADH_2$ and 1 ATP molecules are produced per cycle in krebs cycle.
(1) If both A and R are true and R is the correct explanation of A.
(2) If both A and R are true but R is not the correct explanation of A.
(3) If A is true, but R is false.
(4) If A is false, but R is true.

19. Read the following statements and choose the set of **correct** statements?\.
I. All living organisms need energy for carrying out daily life activities like movement, transport, reproduction, etc. However, the process of exchange of gases in humans, does not require energy as there is not work done in it.
II. During oxidation within a cell, all the energy contained in respiratory substrates is not released free into the cell or in a single step.
III. Plants do not present great demands for gaseous exchanges.
IV. In glycolysis, ATP is consumed at two steps, First, in the conversion of glucose to glucose-6-phophate and second in the conversion of fructose-6-phophate to fructose 1,6-bisphosphate.
V. Conversion of BPGA to PGA is the only energy yielding step.
(1) II, III and IV (2) I and V (3) Only V (4) II and IV

20. Match the following columns.

	Column-I		**Column-II**
A.	RQ	1.	Chemiosmotic ATP synthesis
B.	ETS	2.	Fatigue
C.	Cytochromes	3.	Inner mitochondrial membrane
D.	Lactic acid	4.	Alcoholic fermentation
E.	Yeast	5.	Respirometer

Codes :

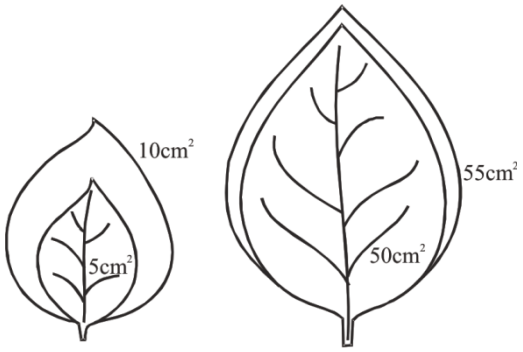
	A	B	C	D	E
(1)	5	1	3	4	2
(2)	5	1	3	2	4
(3)	1	5	2	3	4
(4)	5	2	4	3	1

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**3**	**4**	**1**	**3**	**2**	**2**	**4**	**3**	**1**	**1**	**2**	**1**	**3**	**4**
Que.	16	17	18	19	20										
Ans.	**4**	**2**	**2**	**1**	**2**										

PLANT GROWTH & DEVELOPMENT

1. The given figures shows the growth of two leaves in a given time. If, AGR = absolute growth rate and RGR = relative growth rate, then select the **correct** option :-



B A

	AGR for leaf A	RGR for leaf A	AGR for leaf B	RGR for leaf B
(1)	5	5%	5	5%
(2)	10	10%	55	10%
(3)	5	100 %	5	10 %
(4)	5	10 %	5	100 %

2. Match the following :-

List-I		List-II	
A	Auxin	i	Ripening and maturity of fruits
B	Gibberellin	ii	Differentiation of xylem elements
C	Cytokinin	iii	Speed up malting process
D	Ethylene	iv	Overcome the apical dominance

(1) **A**-iv, **B**-iii, **C**-ii, **D**-i (2) **A**-ii, **B**-iii, **C**-i, **D**-iv

(3) **A**-ii, **B**-iii, **C**-iv, **D**-i (4) **A**-iii, **B**-i, **C**-iv, **D**-ii

3. Exponential growth equation is '$W_1 = W_0 e^{rt}$'. In this equation 'r' is :-

(1) Relative growth rate and is also the measure of efficiency index of the plant

(2) Base of natural logarithms

(3) Time of growth

(4) Initial size

4. How many of the following functions/effects are related to cytokinin ?

Bolting, Lateral shoot growth, Nutrient Mobilisation, Promotes Apical dominance, Closure of Stomata, Delay of leaf senescence, Breaking of seed dormancy

(1) 2 (2) 1 (3) 4 (4) 3

5. **Statement-I :** Growth is regarded as one of the fundamental and conspicuous characteristics of living being.

Statement-II : Growth can be defined as an irreversible permanent increase in size of an organ or its part or even of an individual cell.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Statement-I & Statement-II both are incorrect.

6. Increased vacuolation, cell enlargement and new cell wall deposition are the characteristics of the cells in which of the following phases ?

(1) Meristematic phase of growth (2) Maturation phase of growth

(3) Elongation phase of growth (4) Senescent phase of growth

7. **Assertion:** The final structure at maturity of a cell or tissue is determined by the location of the cell within the plant body.

Reason: Within the plant body the cell or tissues arising out of the same meristem have different structure at maturity.

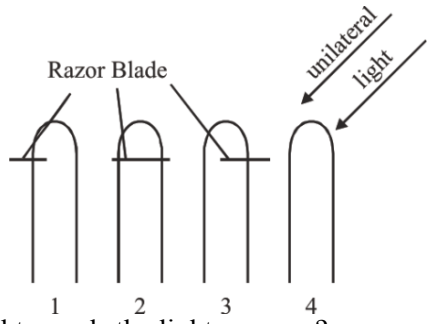
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

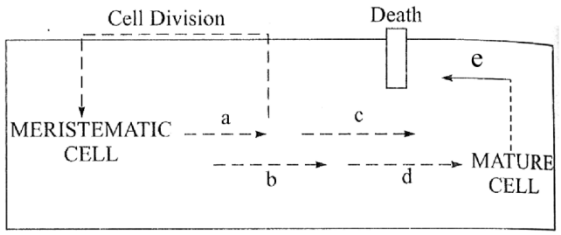
8. The following diagram shows four coleoptiles setup at the start of an experiment.



Which two coleoptiles will bend towards the light sources ?

(1) 1 and 2 (2) 1 and 4 (3) 2 and 3 (4) 3 and 4

9. Recognise the figure and find out the correct labelling.



(1) **a**-elongation, **c**-differentiation, **b**-plasmatic growth, **d**-senescence, **e**-maturation

(2) **b**-elongation, **e**-differentiation, **a**-plasmatic growth, **c**-senescence, **d**-maturation

(3) **a**-elongation, **d**-differentiation, **b**-plasmatic growth, **e**-senescence, **c**-maturation

(4) **b**-elongation, **c**-differentiation, **a**-plasmatic growth, **e**-senescence, **d**-maturation

10. Consider the following four statements **(a-d)** and select the option which includes all the **incorrect** ones only :-

(a) PGR, which is used to speed up the malting process, also promotes internode elongation just prior to flowering in cabbage

(b) PGR, which is called stress hormone, acts as an general plant growth inhibitor.

(c) In most higher plants, the growing apical bud inhibits the growth of the axillary bud. This phenomenon is called decapitation.

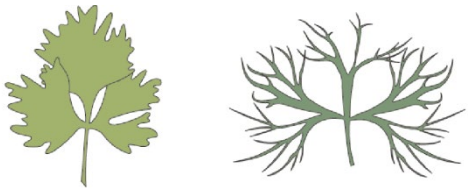
(d) PGR ethylene could fit either of the groups (promoter or inhibitor), but it is largely a promoter of growth activities

Options :-

(1) Statements (b), (c) and (d) (2) Statements (a), (b)

(3) Statements (c), (d) (4) Statements (a), (c) and (d)

11. The following figure shows



Terrestrial habitat Water habitat

(1) Heterophylly in larkspur (2) Heterophylly in buttercup

(3) Heterophylly in cotton and coriander (4) Heterophylly in cotton, coriander and larkspur.

12. Meristem _C_ parenchyma _A_ interfascicular cambium _B_ secondary xylem. Respectively A, B, C refer to :

(1) Differentiation, dedifferentiation and redifferentiation

(2) Dedifferentiation, differentiation and redifferentiation

(3) Differentiation, redifferentiation and dedifferentiation

(4) Dedifferentiation, redifferentiation, differentiation

13. Match the following with respect to discovery of plant growth regulators :-

A	Canary grass	i	Ethylene
B	Bakane of rice	ii	Auxin
C	Tobacco stem callus	iii	GA
D	Ripened orange	iv	Cytokinins

(1) **A**-iv, **B**-iii, **C**-ii, **D**-i (2) **A**-ii, **B**-iii, **C**-iv, **D**-i

(3) **A**-iii, **B**-ii, **C**-iv, **D**-i (4) **A**-ii, **B**-iv, **C**-iii, **D**-i

14. After mitotic division in a cell if one daughter cell continues to divide while the other matures, the formula can be expressed as :

(1) $W_1 = W_0e^{rt}$ (2) $L_1 = L_0e^{rt}$ (3) $W_t = W_0 + rt$ (4) $W_0 = W_1e^{rt}$

15. How many of following functions are **not** related to Gibberellins ?

Internode elongation, malting, bolting, promotes abscission, Sprouting of potato tubers, increasing yield of sugarcane, promote seed dormancy.

(1) 2 (2) 3 (3) 4 (4) 5

16. **Statement-I :** In plant growth, the role of PGR is of only one kind of extrinsic control.

Statement-II : Along with genomic control and extrinsic factors, PGR play an important role in plant growth and development.

(1) Statement-I and II both are correct. (2) Statement-I and II both are incorrect.

(3) Only Statement-I is correct. (4) Only Statement-II is correct.

17. **Assertion :** Development is a term that includes all changes that an organism goes through during its life cycle.
Reason : Plants follows different pathways in response to environment or phases of life to form different kinds of structures.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

18. **Assertion :** In arithmetic growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures.
Reason : A linear curve is obtained when any growth parameter is plotted against time for arithmetic growth.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true and reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

19. Represented below are the two phenomena in plants. Among the following two phenomena **(a)** and **(b)** which phenomena is widely applied in tea plantation?



(1) Phenomena (a)
(2) Phenomena (b)
(3) Phenomena (a) and (b) both
(4) Neither phenomena (a) nor (b)

20. The PGRs can be broadly divided into two groups. One group of PGRs is involved in growth promoting activities and the PGRs of other group :-
(A) play an important role in plant responses to wounds
(B) are involved in various activities such as dormancy and abscission
(C) play an important role in plant responses to stresses of biotic and abiotic origin
(1) Only A & B are correct
(2) Only B & C are correct
(3) Only A & C are correct
(4) A, B & C are correct

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**3**	**1**	**4**	**1**	**3**	**2**	**4**	**4**	**3**	**2**	**4**	**2**	**3**	**2**
Que.	16	17	18	19	20										
Ans.	**4**	**2**	**2**	**2**	**4**										

BREATHING EXCHANGE OF GASES

1. Volume of air a person can inspire or expire per minute is -

(1) 2500 – 3000 ml (2) 1000 – 1100 ml

(3) 6000 – 8000 ml (4) 1100 – 1200 ml

2. Pneumotaxic centre when sends strong signal then which of the following will not occur ?

(1) Increase in rate of breathing

(2) Decrease in duration of inspiration

(3) Complete filling of lungs

(4) Decrease in duration of expiration

3. The amount of air that cannot be expelled is .....

(1) total lung capacity (2) expiratory reserve volume

(3) functional residual capacity (4) residual volume

4. Trachea is straight tube extending up to the _____ :-

(1) end of thoracic cavity (2) Mid thoracic cavity

(3) end of the neck (4) middle of the neck

5. Conducting part of the respiratory system is formed

(1) From external nostrils up to the initial bronchioles

(2) From external nostrils up to the terminal bronchioles

(3) From external nostrils up to the alveoli

(4) From external nostrils up to the alveoli

6. Maximum volume of O_2 is transported as __A__ and of CO_2 is transported as __B__.

Select the option, which correctly fills A and B :-

(1) A-Oxyhaemoglobin, B-Carboxyhaemoglobin

(2) A-Dissolved in plasma, B – HCO_3^- ion

(3) A-Oxyhaemoglobin, B – HCO_3^- ions

(4) A-Oxyhaemoglobin, B-Carbaminohaemoglobin

7. Binding of oxygen with haemoglobin is primarily related to :

(1) Partial pressure of CO_2 (2) Hydrogen ion concentration

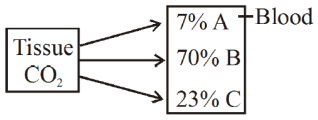
(3) Partial pressure of O_2 (4) Temperature

8. After taking a long deep breath we do not respire for few seconds due to

(1) more CO_2 in blood (2) more O_2 in blood

(3) less CO_2 in blood (4) less O_2 in blood

9.



(1) A-Carboxy compound, B-Carbamino, C-Dissolved form

(2) A-Dissolved form, B-Bicarbonate, C-Carboxy compound

(3) A-Carbamino form, B-Dissolved, C-Carboxy compound

(4) A-Dissolved form, B-Bicarbonate, C-Carbamino compound

10. Which is correct :-

IC $\rightarrow$ Inspiratory capacity

EC $\rightarrow$ Expiratory capacity

FRC $\rightarrow$ Functional Residual capacity

VC $\rightarrow$ Vital capacity

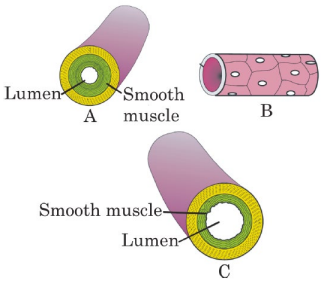
IRV $\rightarrow$ Inspiratory reserve volume

(1) FRC > IC (2) EC > IC (3) VC > FRC (4) IRV > IC

ANSWER KEY										
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**3**	**3**	**4**	**2**	**2**	**3**	**3**	**3**	**4**	**3**

BODY FLUIDS AND CIRCULATION

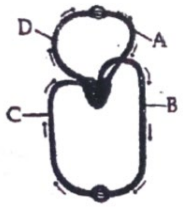
1. Identify A, B and C in the given diagram



Choose the correct option :

(1) A- Artery, B-Capillary, C- Vein
(2) A-Artery, B-Vein, C-Capillary
(3) A-Vein, B-Artery, C-Capillary
(4) A-Capillary, B-Artery, C-Vein

2. Figure shows schematic plan of blood circulation in humans with labels A to D. Identify the correct label and its function/s :-



(1) D-Dorsal aorta - takes blood from heart to body parts, PO_2 = 95 mm Hg
(2) A-Pulmonary vein - takes deoxygenated blood from body, PO_2 = 60mm Hg
(3) B-Dorsal aorta - carry blood from heart to lungs, PO_2 = 45 mm Hg
(4) C-Vena cava - takes de-oxygenated blood from body parts to right auricle, PCO_2 = 45 mm Hg

3. Which of the following is correct option for Blood groups ?

	Blood Group	**Antigens on RBCs**	**Antibodies in plasma**
(1)	A	A	Anti-B
(2)	B	A	Anti-A
(3)	AB	Nil	Anti-A,B
(4)	O	A,B	Nil

4. In ECG the "P" wave represents :-

(1) Depolarisation of the atria
(2) Depolarisation on ventricles
(3) The return of the ventricles from excited state to normal state
(4) Contraction of ventricles

5. High blood pressure damages which set of organs?

(1) Liver, Heart, Kidneys
(2) Liver, Kidneys
(3) Liver, Kidneys and Lungs
(4) Brain, Kidneys, Heart

6. Match the following columns and choose the correct answer from the options given below -

	Column 'A'		Column 'B'
(A)	Bicuspid valve	(i)	between right atrium and right ventricle
(B)	Aortic semilunar valve	(ii)	between left atrium and left ventricle
(C)	Tricuspid valve	(iii)	between right ventricle and pulmonary artery
(D)	Pulmonary semilunar valve	(iv)	Between left ventricle and aorta

	A	B	C	D
(1)	ii	iv	iii	i
(2)	ii	iv	i	iii
(3)	i	ii	iii	iv
(4)	ii	iii	iv	i

7. Blood vessels that contain deoxygenated blood are :-

(1) Systemic arteries and pulmonary vein

(2) Systemic veins and pulmonary arteries

(3) Systemic arteries and pulmonary arteries

(4) Systemic veins and pulmonary veins

8. During high blood pressure, regulations of heart beat and circulation are controlled by -

(1) Vasodilator and vasconstrictor centres

(2) Cardio-stimulatory and vasoconstrictor centres

(3) Cardio-inhibitory and vasoconstrictor centres

(4) Cardio-inhibitory and vasodilator centers.

9. Time duration between DUP and LUBB sound is :

(1) 0.3 seconds (2) 0.1 seconds

(3) 0.5 seconds (4) 0.8 seconds

10. If end diastolic volume is X, Stroke volume is Y and end systolic volume is Z then find out correct formula to calculate ejection fraction?

(1) Ejection fraction = Y/X (2) Ejection fraction = Y/Z

(3) Ejection fraction = X/Y (4) Ejection fraction = Z/Y

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**1**	**4**	**1**	**1**	**4**	**2**	**2**	**4**	**3**	**1**

EXCRETORY PRODUCTS & THEIR ELIMINATION

1. Which of the following blood vessels consist of least amount of urea?
(1) renal artery (2) renal vein (3) hepatic artery (4) hepatic vein

2. The condition of accumulation of urea in the blood is termed as :
(1) Renal calculi (2) Glomerulonephritis (3) Uremia (4) Ketonuria

3. Mark the odd one out :-
(1) Afferent arteriole (2) Vasa-recta (3) Henle's loop (4) Efferent arteriole

4. Which one is true ?
(1) Small amount of nitrogenous wastes could be eliminated through saliva
(2) Sweat produced by sweat gland is watery fluid containing substance like sterols, hydrocarbons and waxes
(3) Presence of glucose and ketone bodies in urine is indicative of Diabetes insipidus
(4) Malfunctioning of kidney can lead to accumulation of amino acid in blood

5. The main function of glomerulus is :-
(1) Filtration of blood (2) Reabsorption of H_2O
(3) Reabsorption of Na^+ (4) Concentration of urine

6. Which of the following group of animals are ureotelic in nature?
(1) Human, tadpole of frog (2) Human, frog
(3) Crustaceans, frog (4) Spider, earthworm

7. You and your study partner want to draw the pathway that controls the reabsorption of sodium ion during hypotension.
Which of the following is the correct sequence of this event ?
(i) Aldosterone is released (ii) Kidney tubules reabsorb Na^+
(iii) Renin is released (iv) JGA recognizes a drop in BP
(v) Angio tensin II is produced
(1) i – iii – v – ii – iv (2) iv – ii – iii – i – v
(3) iv – iii – v – i – ii (4) ii – iv – iii – i – v

8. Our lungs remove CO_2, which is :-
(1) 200 ml/min (2) 400 ml/min (3) 300 ml/min (4) 600 ml/min

9. Conversion of ammonia to urea is done by....cycle :-
(1) Ornithine (2) Arginine (3) Fumaric (4) Citrulline

10. Duct of Bellini present in :-
(1) cortex of kidney (2) papilla of medulla
(3) at junction of cortex and medulla (4) superficial in medulla

ANSWER KEY										
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**2**	**3**	**3**	**1**	**1**	**2**	**3**	**1**	**1**	**2**

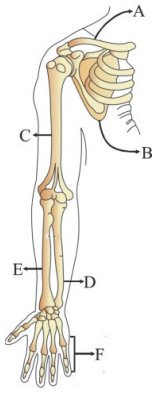
MOVEMENT AND LOCOMOTION (MUSCLES & SKELETAL SYSTEM)

1. Read the following statements and find out the incorrect statement :-
(1) Sliding filament theory states that contraction of a muscle fibre takes place by the sliding of the thick filaments over the thin filaments
(2) Muscle contraction is initiated by a signal sent by the CNS via a sensory neuron
(3) A Sensory neuron along with the muscle fibre connected to it constitute a motor-end plate
(4) All of the above

2. Which statements are incorrect ?
(A) During muscle contraction A-band decreases in length.
(B) Red muscle fibers contains high quantity of myoglobin.
(C) Z-line bisects the I-band.
(D) Length of Myofilament decrease during muscle contraction.
(1) A and D (2) B and D
(3) A and C (4) C and D

3. Name the type of joint between following :-
(A) Between atlas and axis _____?
(B) Between carpals_____?
(C) Between cranial bone_____?
(D) Between femur and Acetabulum cavity_____?
Choose the correct option :
(1) Pivot, Hinge, Suture, Ball and Socket (2) Pivot, Elipsoid, Fibrous, Gliding
(3) Pivot, Gliding, Suture, Ball and Socket (4) Gliding, Pivot, Suture, Ball and Socket

4. How many of the following statements are correct regarding given diagram ?



(a) 'A' is commonly called collar bone and has two curvatures.
(b) 'B' is ventrally situated, large, Triangular, flat bone
(c) Joint formed between C, D and E is a type of Synovial joint
(d) F is larger sesamoid bone
(1) Three (2) Two (3) Four (4) One

5. Match the column–I with column–II and choose the correct option.

	Column-I		**Column-II**
(a)	Myosin	(i)	Troponin-C
(b)	Z-disc	(ii)	Intercalated Disc
(c)	Actin	(iii)	ATPase Enzyme
(d)	Cardiac muscle	(iv)	Actinin Protein

(1) a-iii, b-iv, c-i, d-ii (2) a-i, b-iv, c-ii, d-iii

(3) a-iii, b-ii, c-iv, d-i (4) a-iv, b-i, c-iii, d-ii

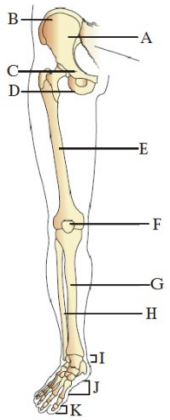
6. Match Column-I with Column-II and select the correct option from the option given below :

Column-I		**Column-II**	
(A)	Hinge joint	(i)	Wrist joint
(B)	Pivot joint	(ii)	Interphalangeal joint
(C)	Ellipsoid joint	(iii)	Radioulnar joint
(D)	Saddle joint	(iv)	First carpometacarpal joint

(1) A-ii; B-i; C-iv; D-iii (2) A-ii; B-iii; C-I; D-iv

(3) A-i; B-ii; C-iv; D-iii (4) A-iii; B-i; C-iv; D-ii

7. Given below is the diagramatic frontal view of right pelvic girdle and lower limb bones :-



(a) 'A' is formed by fusion of 'B', 'C' and 'D'

(b) 'F' is the largest sesamoid bone of body

(c) 'G' does not participates in knee joint

(d) Digital formula comes out to be 23333 when 'K' bones are counted lateral to medial

(e) Obturator foramen is found between (C) and (D)

How many statements is/are correct ?

(1) Two (2) Three (3) Four (4) One

8. Correctly match column I with column II :-

Column I		Column II	
(A)	Tetany	i	Endomysium
(B)	Muscle fibre	ii	Rapid spasm in muscles
(C)	Fasciculi	iii	Auto immune Disorder
(D)	Myasthenia gravis	iv	Perimysium

(1) A→(iii), B→(ii), C→(iv), D→(i)
(2) A→(ii), B→(i), C→(iii), D→(iv)
(3) A→(ii), B→(i), C→(iv), D→(iii)
(4) A→(iii), B→(iv), C→(i), D→(ii)

9. Read the following statements carefully :-

(a) Cross-bridge formation
(b) Neurotransmitter released
(c) Action potential generated in sarcolemma
(d) Shortening of sarcomere
(e) Binding of calcium at Troponin C-site.

Arrange the following statements in a correct sequence of muscle contraction.

(1) c → b → e → a → d
(2) b → c → e → a → d
(3) c → b → a → e → d
(4) a → c → b → d → e

10. Which joint is similar to elbow joint ?

(1) Joint between humerus and scapula
(2) Joint between atlas and axis
(3) Joint between Femur, Tibia and patella
(4) Joint between Radius and Ulna

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**4**	**1**	**3**	**2**	**1**	**2**	**2**	**3**	**2**	**3**

NEURAL CONTROL & CO-ORDINATION

1. Which of the following factor is responsible for resting membrane potential?
(1) more permeability of membrane for Na^+
(2) more permeability of membrane for K^+
(3) presence of negatively charged protein molecule outside the neuron
(4) presence of positively charged protein molecule on inner side of the neuron

2. Hypothalamus does not control :-
(1) Body temperature (2) Urge for eating
(3) Urge for drinking (4) Voluntary movements

3. Opaque white appearance of white matter is due to:-
(1) Fibers of tract covered by axolemma only. (2) Fibers of tract not covered by myelin sheath
(3) Fibers of tract covered by myelin sheath. (4) White matter absent in CNS.

4. In following list how many structures are included in grey matter ?
Cell body, Telodendria, Corpus callosum, Myelinated axon, Unmyelinated axon, Dendron.
(1) Five (2) Three (3) Two (4) Four

5. Which one of the following statements is incorrect ?
(1) At electrical synapses, the membranes of pre and post synaptic neurons are in very close proximity.
(2) Electrical current can flow directly from one neuron into the other across electrical synapse.
(3) Impulse transmission across a chemical synapse is always faster than that across an electrical synapse.
(4) Electrical synapses are rare in our body.

6. During repolarisation :-
(1) Na^+VGC close and K^+ VGC open (2) K^+ VGC close and Na^+ VGC open
(3) K^+ VGC open and Na^+ VGC open (4) Both K^+ VGC and Na^+ VGC close

7. Which part of brain has very convoluted surface in order to provide additional space for many more neurons?
(1) Mid brain (2) Cerebellum (3) Medulla (4) Pons

8. Mark the incorrect statement regarding your mid-brain?
(1) Located between the thalamus/hypothalamus and pons
(2) Receives auditory, tactile and visual inputs
(3) Ventral portion consists of four round swellings (lobes).
(4) Superior lobes are related with visual or pupillary reflexes

9. Read the following statements carefully and find out the incorrect one :-
(1) Well developed neural organisation is found in vertebrates.
(2) Simple neural organisation with nerve net found in hydra.
(3) Better neural organisation is found in insecta
(4) Neural organisation is first developed in insecta

10. In the given list how many structures are not present in forebrain?
Corpora quadrigemina, amygdala, hippocampus, corpus callosum, cerebral aqueduct
(1) 3 (2) 1 (3) 2 (4) 4

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**2**	**4**	**3**	**4**	**3**	**1**	**2**	**3**	**4**	**3**

CHEMICAL COORDINATION AND INTEGRATION

1. Leydig cells produce group of hormones called ?
(1) Androgens (2) Estrogen (3) Aldosterone (4) Gonadotropins

2. Which of the following gland help in the metabolism, pigmentation, the menstrual cycle as well as defense capability :-
(1) Pineal gland (2) Pituitary gland (3) Thyroid gland (4) Thymus gland

3. Which hormone stimulates secretion of pancreatic enzymes and bile juice ?
(1) GIP (2) CCK (3) Gastrin (4) Secretin

4. Which of the following will not increase blood pressure ?
(1) ADH (2) ANF (3) Aldosterone (4) RAAS

5. Which of the following conditions will stimulate parathyroid gland to release parathyroid hormone?
(1) Fall in active Vitamin D levels (2) Fall in blood Ca^{+2} levels
(3) Fall in bone Ca^{+2} levels (4) Rise in blood Ca^{+2} levels

6. Which of the following pairs of hormones are the examples of those, that can easily pass through the cell membrane :
(1) Somatostatin, oxytocin (2) Cortisol, testosterone
(3) Insulin, glucagon (4) Thyroxine, MSH

7. Thymosin is responsible for ?
(1) Raising the blood sugar level (2) Raising the blood calcium level
(3) Differentiation of T lymphocytes (4) Decrease in blood RBC

8. Which hormone work on exocrine pancreas and stimulate secretion of water and bicarbonate ions:-
(1) Gastrin (2) Secretin (3) CCK (4) GIP

9. Which of the following hormone is not proteinaceous in nature ?
(1) TSH (2) Aldosterone (3) LH (4) FSH

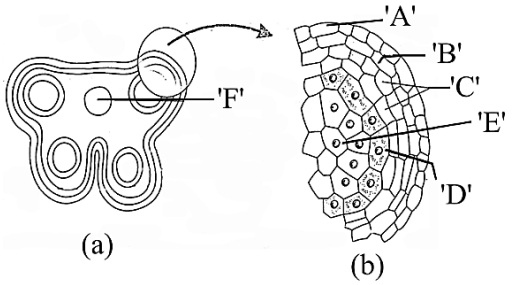
10. Blood Ca^{++} level is regulated by :-
(1) TCT (2) Parathormone (3) Aldosterone (4) Both (1) and (2)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**1**	**1**	**2**	**2**	**2**	**2**	**3**	**2**	**2**	**4**

SEXUAL REPRODUCTION IN FLOWERING PLANTS

1. Given figure 'a' is transverse section of anther and diagram 'b' is enlarged view of one microsporangium showing wall layers. In which labelled structures A to F are :



(1) A → Epidermis, B → Exothecium,
C → Middle layer, D → Tapetum,
E → Microspore mother cells,
F → Connective

(2) A → Epidermis, B → Endothecium,
C → Middle layer, D → Tapetum,
E → Microspore mother cells,
F → Connective

(3) A → Endothecium, B → Epidermis,
C → Tapetum, D → Middle layer,
E → Connective,
F → Microspore mother cells

(4) A → Middle layer, B → Epidermis,
C → Endothecium, D → Tapetum,
E → Microspore mother cells,
F → Connective

2. Read the following four statements (A-D) :-

(A) The site of meiotic division in higher plants is spore mother cell.

(B) In the anther, middle layers lie between endothecium and tapetum.

(C) 2 male gametes are formed from generative cell and vegetative cell.

(D) Pollen grain intine has prominent apertures called germ pores.

How many of the above statements are right :-

(1) Four (2) One (3) Two (4) Three

3. Which of the following statements about sporopollenin is false ?

(1) No enzyme that degrades sporopollenin so far is known.

(2) Sporopollenin is one of the most resistant inorganic material.

(3) Sporopollenin can withstand high temperature and strong acids.

(4) Germpores are the areas where sporopollenin is not present.

4. Given below are two statements :

Assertion (A) :- In over 60 percent of angiosperms, pollen grains are shed at 3-celled stage.

Reason (R) :- The mature pollen grain contains two cells, the vegetative cell and generative cell.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

5. Match the Column-I with Column-II and select the correct option from the codes given below :-

	Column-I		**Column-II**
A	Funicle	(i)	Mass of parenchymatous cells
B	Hilum	(ii)	Basal part of ovule
C	Integument	(iii)	One or two protective envelops of ovule
D	Chalaza	(iv)	Region where body of ovule fuses with funicle
E	Nucellus	(v)	Stalk of ovule

(1) A-(i), B-(ii), C-(iii), D-(iv), E-(v)

(2) A-(v), B-(iv), C-(iii), D-(ii), E-(i)

(3) A-(iv), B-(ii), C-(i), D-(iii), E-(v)

(4) A-(i), B-(iii), C-(v), D-(ii), E-(iv)

6. Select the incorrect statement from the following -

(1) Wind pollination is quite common in grasses.

(2) In majority of aquatic plants such as water hyacinth and water lily, pollination occurs by insect or wind.

(3) Wind pollinated flowers often have a single ovule in each ovary

(4) Triple fusion involves the fusion of two synergids and one male gamete

7. Identify the set of correct statements :

A. The flowers of *Vallisneria* are colourful and produce nectar.

B. The flowers of waterlily are not pollinated by water.

C. In most of water-pollinated species, the pollen grains are protected from wetting by mucillage.

D. Pollen grains of some hydrophytes are long and ribbon like.

E. In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below :

(1) C, D and E only (2) A, B, C and D only

(3) A, C, D and E only (4) B, C, D and E only

8. How many of the following statements are correct ?
(a) Scutellum is found in monocot embryos.
(b) Pollination and fertilisation are fully dependent on water in angiosperms.
(c) Integument cells protrude into embryo sac and develop into the embryos in citrus and mango.
(d) If hybrid are made into apomicts, there is no segregation of characters in the hybrid progeny.
(1) Four (2) Three (3) One (4) Two

9. Dicot embryo consists of :
(1) Radicle and plumule only
(2) Radicle, plumule, and two cotyledons
(3) Radicle, plumule, two cotyledons and tegmen
(4) Radicle, plumule, two cotyledons, tegmen and testa

10. Which one of the following options gives the correct categorisation of plants according to the type of pollination (A, B, C) they have :-

	Anemophily	**Entomophily**	**Hydrophily**
(1)	*Hydrilla, Zostera*	Grass, Maize	*Yucca, Amorphophallus*
(2)	Grass, Maize	*Hydrilla, Zostera*	*Yucca, Amorphophallus*
(3)	Grass, Maize	*Yucca, AmorphophalIus*	*Hydrilla, Zostera*
(4)	*Yucca, Amorphophallus*	Grass, Maize	*Hydrilla, Zostera*

11. Which of the following statement for flower is wrong?
(1) Morphological and embryological marvels in angiosperms.
(2) Sites of sexual reproduction.
(3) Objects of aesthetic, ornamental, social, religious and cultural value.
(4) Calyx and gynoecium are essential whorls.

12. Which of the following is/are not correct with respect to megasporogenesis and megagametogenesis ?
(A) Ovules generally differentiate a group of megaspore mother cells in the micropylar region of the nucellus.
(B) In a majority of flowering plants, one of the megaspore is functional.
(C) During the formation of 8 nucleated structure mitotic divisions are strictly free nuclear initially.
(D) Synergids have special cellular thickenings at the chalazal tip, they are called filiform apparatus.
(1) A and C (2) B and D
(3) B and C (4) A and D

13. In an angiosperms, how many microspore mother cells are required to produce 100 pollen grains?
(1) 50 (2) 75 (3) 25 (4) 100

14. What is correct about the pollen grain of flowering plants ?
(1) Large vegetative cell with large nucleus
(2) Small vegetative cell with small nucleus
(3) Small vegetative cell with large nucleus
(4) Small generative cell with large nucleus

15. **Statement I :-** Pollen grains are rich in nutrients.

Statement II :- In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release.

(1) Statement-I and statement-II both are true

(2) Statement-I is true and statement-II is false

(3) Statement-I is false and statement-II is true

(4) Statement-I and statement-II both are false

16. Which mineral element is mainly needed for pollen tube formation ?

(1) Bromine (2) Boron (3) Calcium (4) Iron

17. **Assertion :-** In monoecious plants neither autogamy nor geitonogamy can be observed.

Reason :- In monoecious plants male and female flowers are produced on different plants.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion

(3) Assertion is True but the Reason is False

(4) Both Assertion & Reason are False

18. Consider the following features seen in a plant :
I. Male and female reproductive organs are generally found in separate flowers.
II. The male flowers having a number of long filaments and have exposed stamens.
III. The female flowers having long, feather like stigmas.
The flowers of this plant would most likely be pollinated by :
(1) Wind (2) Water (3) Bees (4) Birds

19. In which of the following plants the seeds have no residual endosperm as it is completely consumed during embryo development ?
(1) Pea & ground nut (2) Barley and castor
(3) Castor and ground nut (4) Maize and wheat

20. Which of the following statement is incorrect ?
(1) Pollen grains of many species can germinate on stigma of a flower, but only one pollen tube of same species grows into style
(2) Pollen germination and pollen tube growth are regulated by chemical components of pollen interacting with those of pistil
(3) Insect that consume pollen or nectar without bringing out pollination are called pollen/nectar robbers
(4) Some reptiles have been reported as pollinators in some plant species

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**3**	**2**	**3**	**2**	**4**	**4**	**4**	**2**	**3**	**4**	**4**	**3**	**1**	**1**
Que.	16	17	18	19	20										
Ans.	**2**	**4**	**1**	**1**	**1**										

HUMAN REPRODUCTION

1. Which of the following is injected by the doctors to induce delivery in pregnant woman?
(1) Progestrone (2) Thyroxine
(3) Relaxin (4) Oxytocin

2. Select the correct option of haploid cells from the following groups :-
(1) Primary oocyte, Secondary oocyte, Spermatid
(2) Secondary spermatocyte, First polar body, Ovum
(3) Spermatogonia, Primary spermatocyte, Spermatid
(4) Primary spermatocyte, Secondary spermatocyte, Second polar body

3. Which of the following is not a significance of fertilization ?
(1) Completion of oogenesis
(2) Formation of germ layers
(3) Restoration of normal diploid number of chromosomes
(4) Induction of fertilised egg to undergo cleavage due to entry of proximal centriole of sperm into egg itself

4. Which of the following hormone is responsible for controlling levels of FSH ?
(1) Relaxin and oxytocin (2) Estrogen and inhibin
(3) Relaxin and testosterone (4) Inhibin and relaxin

5. Cells produced by 1st meiotic division of spermatogenesis are :-
(1) Structurally and genetically similar to each other
(2) Structurally and genetically dissimilar to each other
(3) Structurally similar but genetically dissimilar to each other
(4) Structurally dissimilar and genetically similar to each other

6. If mammalian ovum fails to get fertilised, which one of the following is unlikely ?
(1) Corpus luteum will degenerate (2) Estrogen secretion further decreases
(3) Primary follicle starts ovary will degenerate (4) Progesterone secretion rapidly declines

7. **Statement I :-** After implantation finger like projections appear on inner cell mass called chorionic villi
Statement II :- In the later stage of pregnancy a hormone called hCG is also secreted by the ovary.
Choose the correct answer from the given options :-
(1) Both Statement I and Statement II are incorrect
(2) Both Statement I and Statement II are correct
(3) Statement I is correct but Statement II is not correct
(4) Statement I is incorrect but Statement II is correct

8. **Statement-I :** When blastocyst gets embedded in endometrium, it is called as implantation.
Statement-II : Blastula has same amount of cytoplasm but much more DNA than zygote.
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is true and Statement-II is false.
(4) Statement-I is false and Statement-II is true.

9. If cowper's gland is removed which of the following will be affected
(1) Sexual attraction (2) Capacitation of sperms
(3) Hardness of penis (4) Copulation and fertilization

10. Nebenkern is :-
(1) Sheath of cytoplasm (2) Acrosome membrane
(3) Mitochondrial arrangement (4) Nuclear membrane

11. Which one of the following is the correct matching of the events occurring during menstrual cycle?
(1) Ovulation-Sharp fall in LH & FSH
(2) Proliferative phase-Rapid regeneration of myometrium and maturation of Graafian follicle
(3) Development of corpus luteum-Secretory phase and increased secretion of progesterone
(4) Menstruation-Breakdown of myometrium and ovum not fertilized

12. In human embryo, the extra embryonic membranes are formed by :-
(1) inner cell mass (2) trophoblast (3) formative cells (4) follicular cells

13. Primary spermatocyte differs from spermatogonium in
(1) Number of autosomal chromosomes (2) Cellular size and volume
(3) Number of sex chromosomes (4) Number of chromosomes

14. Hormone administered for hastening child birth is meant for :-
(1) Stimulating the striped muscles (2) Raising blood pressure
(3) Increasing energy availability (4) Contraction of smooth muscles

15. In which uterine phase does the functional layer of endometrium starts to rebuilt ?
(1) Menstrual phase (2) Secretory phase (3) Follicular phase (4) Bleeding phase

16. The testes in humans are situated outside the abdominal cavity inside a pouch called scrotum. The purpose served is for :
(1) Acceleration of maturation of sperms
(2) Providing more space for the growth of epididymis
(3) Escaping any possible compression by the visceral organs
(4) Maintaining the scrotal temperature lower than the internal body temperature.

17. What is the role of zona pellucida after fertilisation?
(1) Formation of chorionic villi. (2) Formation of germ layers.
(3) To prevent polyspermy. (4) To prevent ectopic pregnancy.

18. **Statement-I :** If Menstrual period is of 40 days, day 22 to 29 of the cycle are called fertile period.
Statement-II : Chance of fertilization are very high during risk period.
(1) Both Statement-I and Statement-II are correct. (2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is true and Statement-II is false. (4) Statement-I is false and Statement-II is true.

19. Which structure is not surrounded by zona pellucida :-
(1) Ovum (2) Secondary Oocyte
(3) Gastrula (4) Morula

20. Most of the non-cellular fluid in ejaculated human semen is composed of :-
(1) The secretion of seminiferous tubules (2) The secretion of bulbourethral glands
(3) The secretion of seminal vesicles (4) The secretion of prostate gland

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**2**	**2**	**2**	**3**	**3**	**1**	**1**	**4**	**3**	**3**	**2**	**2**	**4**	**3**
Que.	16	17	18	19	20										
Ans.	**4**	**4**	**1**	**3**	**3**										

REPRODUCTIVE HEALTH

1. Which is not used as spacing method of birth control :-

(1) Periodic abstinence (2) Condoms

(3) IUDs (4) Vasectomy

2. According to world health organisation, reproductive health means total well - being in

(1) physiological and chemical aspects of reproduction.

(2) structural and organisational aspect of reproduction.

(3) physical, emotional, behavioural and social aspects of reproduction.

(4) sex- related diseases.

3. Match the following STD with its Pathogen :-

(i)	AIDS	(a)	Virus
(ii)	Trichomoniasis	(b)	Protozoa
(iii)	Syphilis	(c)	Bacteria

(1) i-a, ii-b, iii-c (2) i-b, ii-c, iii-a (3) i-c, ii-b, iii-a (4) i-c, ii-a, iii-b

4. Match the column–I with column–II

Column–I		**Column–II**	
A.	Genital warts	(i)	Chlamydia trachomatis
B.	Gonorrhoea	(ii)	Treponema pallidum
C.	Genital Herpes	(iii)	Herpes simplex virus
D.	Syphilis	(iv)	Nisseria gonorrhoeae
		(v)	Human papilloma virus

(1) A–iii, B–iv, C–v, D–ii (2) A–v, B–iv, C–iii, D–ii

(3) A–i, B–iv, C–iii, D–ii (4) A–ii, B–iv, C–i, D–iii

5. Periodic abstinence is a birth control method which includes to avoid the sexual relation during fertile period

(1) Days 10-17 of menstrual cycle (2) Days 01-05 of menstrual cycle

(3) Days 19-22 of menstrual cycle (4) Days 20-25 of menstrual cycle

6. If male is impotent and female is fertile then what possible assisted reproductive technology is used for treating the infertility in particular couple?

(1) GIFT (2) Intrauterine insemination

(3) ZIFT (4) Embryo transfer

7. Population growth can be controlled by all except

(1) Encouraging couples to use contraceptive methods

(2) Giving incentives to couples with smaller families

(3) Raising the marriageable age of males and females

(4) Promoting unprotected sexual co-habitation

8. How many of the following are sexually transmitted diseases ?
AIDS, Trichomoniasis, Syphilis, Chlamydiasis, Genital herpes, diphtheria, tuberculosis
(1) Three (2) Six (3) Four (4) Five

9. Which one of the following pair of disease can spread through blood transfusion.
(1) AIDS and Malaria (2) AIDS and Hepatitis-A
(3) Diabetes mellitus and syphilis (4) Hepatitis B and AIDS

10. How many factors are related to infertility-
(A) Psychological (B) Immunological (C) Physical (D) Congenital
(E) Drugs (F) Diseases
(1) Three (2) Four (3) Five (4) Six

ANSWER KEY											
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	**4**	**3**	**1**	**2**	**1**	**2**	**4**	**4**	**4**	**4**	

PRINCIPLES OF INHERITANCE AND VARIATION

1. Match Column - I with Column- II :

	Column - I		**Column - II**
(a)	Down's syndrome	(i)	11th chromosome
(b)	Alpha -Thalassemia	(ii)	16th chromosome
(c)	Beta -Thalassemia	(iii)	X- chromosome
(d)	Klinefelter's syndrome	(iv)	21st chromosome

(1) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(2) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)

(3) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)

(4) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii)

2. Genotype of plant is +g+h/a+b+ in which all genes show complete linkage. Find out percentage of gamete containing genotype?

(1) 50% (2) 100% (3) 0% (4) 25%

3. Pure yellow pod colour garden pea plant is crossed with pure green pod garden pea plant and produce 1000 offspring in F_2 generation. How many of them have green pod.

(1) 750 (2) 250 (3) 500 (4) 1000

4. Given below are two statements:

Statement I : α Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 11.

Statement II : β Thalassemia is controlled by a single gene HBB on chromosome 16.

In the light of above Statement choose the correct answer from the options given below:

(1) Statement I is incorrect but Statement II is correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Both Statement I and Statement II are correct

5. Given below are two statements one is labelled as **Assertion(A)** and the other is labelled as **Reason(R)**.

Assertion(A) : In honey bee male do not have grandfather and thus cannot have grandsons, but have a father and can have son.

Reason(R) : In honey bee males produce sperms by mitosis.

In the light of above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R)are correct and (R)is the correct explanation of (A)

(2) Both (A) and (R)are correct and (R)is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

6. Match Column-I with Column-II

Column-I	Column-II
(a) Genetics	(i) Passing of characters from parents to progeny
(b) Inheritance	(ii) Laws of inheritance
(c) Variation	(iii) Degree of differences of progeny from their parents.
(d) Mendel	(iv) Branch of biology deals with inheritance and variation.

(1) a–iv; b–iii; c–i; d–ii (2) a–iii; b–iv; c–ii; d–i

(3) a–ii; b–i; c–iv; d–iii (4) a–iv; b–i; c–iii; d–ii

7. **Statement-I :** Law of segregation is based on the fact that the alleles do not show any blending and both the characters are recovered as such in F_1 generation.

Statement-II : Law of dominance is used to explain the expression of only one of parental character in a monohybrid cross in the F_1 and expression of both in the F_2.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct and Statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

8. In a plant black seed colour (BB/Bb) is dominant over white seed colour (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it ?

(1) bb (2) BB (3) Bb (4) BB/Bb

9. If a garden pea produces 2560 seeds in F_2 generation during a dihybrid cross between round-yellow and and wrinkled-green plant. Then how many seed are wrinkled-yellow, round-yellow and wrinkled-green respectively.

(1) 640, 480, 1280 (2) 480, 1440, 160 (3) 640, 1280, 320 (4) 160, 1440, 480

10. **Assertion:** The law of independent assortment can be studied through monohybrid cross.

Reason: When two pairs of traits are combined in a hybrid, segregation of one pair of characters is always dependent of the other pair of characters.

(1) Assertion and reason both are correct and reason is correct explanation for assertion.

(2) Both assertion and reason are incorrect.

(3) Assertion only correct.

(4) Assertion and reason both are correct and reason is not explaining assertion.

11. Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles do not show any blending and both the characters appear as such in F_2 generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called factor.

E. The expression of only one of the parental characters is found in a monohybrid cross in the F_1.

Choose the correct answer from the options given below :

(1) A, B and C only (2) B, C and D only

(3) A, C, D and E only (4) A, B, C, D and E

12. A colour blind male marries a normal female, there is no history of colour blindness in her family. Which of the following is a correct statement

(A) All son will be colourblind

(B) 50% daughter will be colourblind

(C) All son will be normal

(D) All daughter will be carrier

(E) 50% son will be colourblind

(1) Only (A), (B) and (D) (2) Only (B) and (E)

(3) Only (A), (B), (C) and (E) (4) Only (C) and (D)

13. Blood group of father is AB and mother has heterozygous A blood group. which blood group is not possible in children.

(A) A (B) AB (C) O (D) B

(1) Only (A) and (B) (2) Only (C)

(3) Only (B) and (C) (4) Only (A), (B) and (D)

14. With respect to flower position and flower colour genotype of garden pea plant is AaWw. This plant is crossed with AAWw genotypic plant . What phenotypic ratio will be obtained:-

(1) 3:1 (2) 1:2:2:4:1:2:1:2:1

(3) 3:1:3:1 (4) 1:2:1:1:2:1

15. Given below are two statements:

Statement I : Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called aneuploidy.

Statement II : Failure of cytokinesis after telophase stage of cell division results in an increase in a whole set of chromosomes in an organism and, this phenomenon is known as polyploidy.

In the light of above Statement choose the correct answer from the options given below:

(1) Statement I is incorrect but Statement II is correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Both Statement I and Statement II are correct

16. **Assertion :** The phenotype/trait will only be dependent on the functioning of the unmodified allele in the presence of a non-functional enzyme.

Reason : The unmodified allele, which represents the original phenotype, is generally dominant.

(1) Both assertion and reason are true, and reason is the correct explanation of assertion.

(2) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(3) Assertion is true, but reason is false.

(4) Assertion is false, but reason is true.

17. **Statement-I :** The segregation of alleles is a random process and so there is a 50% chance of a gamete containing either alleles.

Statement-II : The factors did not show any bending at either F_1 or F_2.

(1) Both Statement-I and II are correct.

(2) Both Statement-I and II are incorrect.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

18. **Statement-I :** Phenotypic ratio of Monohybrid cross performed by Mendel was 3:1 in F_2 generation.

Statement-II : Dwarf plant recovered in F_2 generation due to segregation of alleles.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

19. In rabbit black skin (B) is dominant over brown skin(b) and short (S) is dominant over long hair(s) if homozygous black-short haired male is crossed with homozygous brown long haired female. All F_1 offspring are heterozygous black short haired F_1 male crossed with F_1 female. If F_2 generation what is the percentage of homozygous black short haired offspring?

(1) 50% (2) 12.5% (3) 6.25% (4) 28%

20. A cross is done between the pea plants having Round and Yellow seed (RRYY) and Wrinkled and Green seed (rryy). Find out the ratio of genotypes RrYY, rrYY, rrYy, RRyy in F2 generation ?

(1) 2 : 1 : 2 :1 (2) 1 : 2 : 4 :1 (3) 1 : 1 : 1 :1 (4) 2 : 2 : 4 :1

ANSWER KEY															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**3**	**1**	**2**	**4**	**4**	**3**	**1**	**2**	**2**	**3**	**4**	**2**	**1**	**4**
Que.	16	17	18	19	20										
Ans.	**1**	**1**	**1**	**3**	**1**										

MOLECULAR BASIS OF INHERITANCE

1. Read the following statements and choose the correct set of statements. With respect to goals of human genome project :-

(a) Identify all the genes in human DNA

(b) Improve tools for data analysis

(c) Determine the sequences of the 3 billion chemical base pairs that make up human DNA;

(d) Transfer related technologies to other sectors, such as industries

Choose the correct answer from the options given below:-

(1) (a),(b) and (d) only (2) (c) and (d) only

(3) (a),(b),(c) and (d) only (4) (a) and (b) only

2. What are mutagens :-

(1) chemical and physical factors that induce crossing over

(2) chemical and physical factors that induce mutations

(3) chemical and physical factors that induce Linkage

(4) chemical and physical factors that induce fusion of two cells

3. Regulation of lac operon is under control of :-

(1) Only negative regulation

(2) Only positive regulation

(3) Both negative and positive regulation

(4) Neither negative nor positive regulation

4. In the presence of low glucose when a mutation occurs in i gene by which repressor protein is formed but its binding with operator is affected, what will be the outcomes :-

(1) z, y, a genes will be transcribed

(2) z, y, a genes will not be translated

(3) RNA polymerase will not bind the promoter region

(4) i gene will not get transcribed

5. In dsDNA adenine pair with thymine and guanine pair with cytosine by how many hydrogen bonds respectively ?

(1) 1, 2 (2) 2, 3 (3) 2, 1 (4) 3, 2

6. An E.coli cell with heavy DNA is incubated in culture medium containing light isotope of nitrogen. After 40 minutes find out percentage of light DNA?

(1) 50% (2) 25%

(3) 12.5% (4) 0%

7. In a typical eukaryotic cell 600 nucleosomes are present. How many N glycosidic bonds are present in the DNA ?

(1) 240000 (2) 120000 (3) 480000 (4) 60000

8. Given below are two statements :

Statement-I :- The promoter and terminator flank the structural gene in a transcription unit.

Statement-II :- The presence of a promoter in a transcription unit that also defines the template and coding strands.

In the light of above Statement choose the correct answer from the options given below :

(1) Both Statement I and Statement II are correct (2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are incorrect (4) Statement I is correct but Statement II is incorrect

9. Given below are two statements :

Statement-I :- The split-gene arrangement further complicates the definition of a gene in terms of a DNA segment.

Statement-II :- Exons are said to be those sequence that appear in mature or processed RNA.

In the light of above Statement choose the correct answer from the options given below :

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

10. Some amino acids are coded by more than one codon, hence the codon is :-

(1) Universal (2) Commaless (3) Ambiguous (4) Degenerate

11. Given below are two statements:

Statement-I :- Repeated sequences make up very large portion of the human genome.

Statement-II :- The functions are unknown for over 98 per cent of the discovered genes.

In the light of above Statement choose the correct answer from the options given below:

(1) Both Statement I and Statement II are incorrect.

(2) Both Statement I and Statement II are correct.

(3) Statement I is incorrect but Statement II is correct.

(4) Statement I is correct but Statement II is incorrect.

12. Given below are two statements:

Statement-I :- Physical map of genome was generated using information on polymorphism of restriction endonuclease recognition sites.

Statement-II :- Some repetitive DNA sequences known as microsatellites also used in mapping of human genome.

In the light of above Statement choose the correct answer from the options given below:

(1) Both Statement I and Statement II are incorrect.

(2) Both Statement I and Statement II are correct.

(3) Statement I is incorrect but Statement II is correct.

(4) Statement I is correct but Statement II is incorrect.

13. Which type of mutation is commonly observed in cancer cells :-

(1) Euploidy (2) Aneuploidy

(3) Chromosomal aberration (4) Substitution

14. Given below are two statements:

Statement-I :- Lac operon is the prototype operon in bacteria.

Statement-II :- The repressor protein binds to the promotor region of the operon and prevents RNA polymerase from transcribing the operon.

In the light of above Statement choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

15. Read the following statements and choose the correct set of statements:

(a) The enzymatic method developed by Har Gobind Khorana was instrumental in synthesising amino acid molecules.

(b) The presence of a catalyst would not enhance the rate of peptide bond formation.

(c) During gene regulation, regulatory proteins can act both positively (activators) and negatively (repressors)

(d) In lac operon polycistronic structural gene is regulated by a common promoter and regulatory genes

Choose the correct answer from the options given below :-

(1) (a),(b) and (d) only (2) (c) and (d) only

(3) (a),(b),(c) and (d) only (4) (a) and (b) only

16. Which of the following statement is correct.

(1) Protein-digesting enzymes (proteases) and RNA-digesting enzymes (RNases) affect transformation

(2) Viruses grown in the presence of radioactive phosphorus contained radioactive protein

(3) Viruses grown on radioactive sulfur contained radioactive DNA

(4) DNA being chemically and structurally more stable is a better genetic material.

17. Read the following statements and choose the correct set of statements :

(a) Nucleic acids are long polymers of nucleotides.

(b) A typical nucleosome contains 200 bp of DNA helix.

(c) DNA and RNA both function as genetic material.

(d) 5-methyl uracil, is another chemical name for thymine

Choose the correct answer from the options given below :-

(1) (a),(b),(c) and (d) only (2) (a),(c) and (d) only (3) (a) and (d) only (4) (a),and (c) only

18. Given below are two statements :

Statement-I : If both strands of DNA act as a template, they would code for RNA molecule with identical sequences.

Statement-II : DNA dependent RNA polymerase catalyse the polymerisation in only one direction, That is 3'-5'.

In the light of above Statement choose the correct answer from the options given below :

(1) Both Statement I and Statement II are correct (2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are incorrect (4) Statement I is correct but Statement II is incorrect

19. Sequence of nitrogen bases in one strand of ds DNA is 5' CTAGGCA 3' what will be sequence of nitrogen base in mRNA formed by this ds DNA?

(1) 5' CAUGGCU 3'
(2) 5' CTAGGCA 3'
(3) 5' CUAGGA 3'
(4) 5' ACGGAUC 3'

20. Given below are two statements :

Statement-I :- DNA has evolved from RNA with chemical modifications

Statement-II :- RNA used to act as a genetic material as well as a catalyst

In the light of above Statement choose the correct answer from the options given below :

(1) Both Statement I and Statement II are incorrect
(2) Both Statement I and Statement II are correct
(3) Statement I is incorrect but Statement II is correct
(4) Statement I is correct but Statement II is incorrect

ANSWER KEY															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**2**	**3**	**1**	**2**	**1**	**1**	**1**	**3**	**4**	**4**	**2**	**3**	**1**	**2**
Que.	16	17	18	19	20										
Ans.	**4**	**1**	**3**	**3**	**2**										

EVOLUTION

1. Match the column-I with column-II and choose the correct option ?

	Column-I		Column-II
(A)	Weismann	(i)	Mutation theory
(B)	Hugo devries	(ii)	Inheritance of acquired character
(C)	Ernst Haeckel	(iii)	Biogenetic law
(D)	Lamarck	(iv)	Theory of germ plasm

(1) A - iii B - i C - iv D - ii (2) A - ii B - iii C - iv D - i

(3) A - iv B - ii C - i D - iii (4) A - iv B - i C - iii D - ii

2. According to Darwin Adaptive ability is inheritated it has a genetic basis. What is genetic basis of adaptation ?

(1) Natural selection (2) Heritable somatic variation

(3) Variations in population (4) Pre existing advantageous gene mutations

3. Earthworm and snake have the same body form as both live in burrows. This type of evolution is called :-

(1) Adaptive radiation (2) Convergent evolution

(3) Divergent evolution (4) Micro evolution

4. The most apparent change during the evolutionary history of Homo sapiens is raced in

(1) Loss of body hair (2) Walking upright

(3) Shortening of the jaws (4) Remarkable increase in the brain size

5. The earliest hominid that evolved more than two million years ago were

(1) Homo erectus (2) Australopithecus

(3) Cro-Magnon Man (4) Neanderthal Man

6. Which of the following statements are correct?

A. Homologous organs are the result of divergent evolution.

B. Analogous organs are the result of convergent evolution.

C. Adaptive radiation is the process of evolution of different species in a given geographical area starting from a point and radiating to other areas of geography.

D. Wings of butterfly and wings of birds are homologous organs.

E. Australian marsupials represent adaptive radiation.

Choose the correct answer :

(1) A, B, C only (2) B, C, D, E only

(3) A, B, C, E only (4) A, D, E only

7. Which of the following options show corrects equence of evolution of plant forms :-

(1) Tracheophyte → Ferns → Bryophytes → Seed ferns

(2) Cycads → Gnetales → Ginkgos → Psilophyton

(3) Rhynia → Psilophyton → Progymnosperms → Seed ferns → Dicotyledons

(4) Lycopods → Conifers → Sphenopsids → Monocotyledons

8. Evolution can be best explained by :

(1) Darwin's concept of Descent with modification

(2) Miller's experiment

(3) Louis Pasteur's Experiment

(4) Special creation theory

9. First living beings evolved on earth.....:-

(1) Chemoautotrophs, Anaerobes (2) Chemoheterotrophs, Anaerobes

(3) Photoheterotrophs, Aerobes (4) Photoautotrophs, Aerobes

10. Which of the following population would most quickly lead to two groups with few shared traits?

(1) A population with stabilizing selection. (2) A population with directional selection.

(3) A population with disruptive selection. (4) A population with no selection.

11. Liberation of oxygen by which bacteria brought a revolutionary change in history of earth ?

(1) Nitrifying bacteria (2) Planktonic sulphur bacteria

(3) Iron bacteria (4) Cyanobacteria

12. Which of the following is not closely related with dinosaurs from evolutionary point of view ?

(1) Crocodiles (2) Archaeopteryx (3) Lizards (4) Pteranodon

13. Arrange correct sequence of geologicalperiods :-

(a) Ordovician (b) Devonian (c) Cambrian (d) Carboniferous

(e) Permian (f) Silurian

(1) $c \rightarrow a \rightarrow f \rightarrow b \rightarrow d \rightarrow e$ (2) $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f$

(3) $b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow a$ (4) $d \rightarrow e \rightarrow f \rightarrow a \rightarrow b \rightarrow c$

14. The sequence of origin of life may be :

(1) Inorganic materials $\rightarrow$ Organic materials $\rightarrow$ Colloidal aggregate $\rightarrow$ Eobiont $\rightarrow$ Cell

(2) Organic materials $\rightarrow$ Inorganic materials $\rightarrow$ Colloidal aggregate $\rightarrow$ Eobiont $\rightarrow$ Cell

(3) Inorganic materials $\rightarrow$ Organic materials $\rightarrow$ Eobiont $\rightarrow$ Cell $\rightarrow$ Colloidalaggregate

(4) Organic materials $\rightarrow$ Inorganic materials $\rightarrow$ Eobiont $\rightarrow$ Cell $\rightarrow$ Colloidalaggregate

15. How many of the following statements arecorrect for gene migration or gene flow :-

(a) Gene frequencies change in the originalpopulation

(b) Gene frequencies change in the newpopulation

(c) New gene/alleles are added to the newpopulation

(d) Genes are lost from the old population

(1) Four (2) Three (3) Two (4) One

ANSWER KEY															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**4**	**2**	**4**	**2**	**3**	**3**	**1**	**2**	**3**	**4**	**3**	**1**	**1**	**1**

HUMAN HEALTH AND DISEASE

1. Hepatitis-B vaccine is prepared using :-

(1) Weakened pathogen (2) Inactivating toxin

(3) Live pathogen (4) Recombinant DNA technology

2. Slowly developing chronic inflammation of the organs, usually lymphatic vessels of the lower limb is indicative of :-

(1) Ascariasis (2) Ring worm (3) Filariasis (4) Amoebiasis

3. Which one of the following combinations acts as a usual antigen binding site of an antibody?

(1) Variable regions of a light and another heavy chain

(2) Variable regions of two light chains

(3) Variable regions of two heavy chains

(4) Variable region of a heavy chain and constant region of a light chain

4. Disease with symptoms-fever, pain in joints, whose causative organism is of virus family and transferred by mosquito (Aedes ageypti). Identify the disease?

(1) Enteric fever (2) Chikungunya (3) Rheumatic arthritis (4) Poliomyelitis

5. Which of the following set include the bacterial disease?

(1) cholera, typhoid, mumps (2) tetanus, tuberculosis, dengue

(3) malaria, mumps, polio (4) diphtheria, typhoid, pneumonia

6. Short-lived immunity acquired from mother to foetus across placenta and through mother's milk to infant is respectively due to

(1) IgG, IgM (2) IgA, IgM (3) IgG, IgA (4) IgG, IgD

7. Select the correct set of correct statements from the options given below.

(A) Innate immunity is non specific.

(B) Bacteria infected cells secrete protein called interferons

(C) the T-lymphocytes mediate CMI

(D) Typhold fever could be confirmed by Widal test.

(1) A & B are correct (2) B & C are correct (3) C & D are correct (4) A, C & D are correct

8. Infective stage of plasmodium for human.

(1) Sporozoite (2) Gametocyte

(3) Haemozoin (4) Spores

9. Choose the correct statement from given below

(a) The acceptance or rejection of a kidney transplant depends on specific passive immunity

(b) Cytotoxic–T–cell is responsible for graft rejection

(c) S.C.I.D. is an autoimmune disease.

(d) Rheumatoid arthritis is caused by IgE type antibodies

(1) One (2) Two (3) Three (4) Four

10. Each immunoglobin has two heavy chains andtwo light chains, the antigen binding site ispresent at ?

(1) Variable region of heavy chain

(2) Variable region of both heavy and lightchain

(3) Variable region of light chain.

(4) Constant region of both light and heavychain.

11. Treatment of snake bite with anti venom is an example of

(1) Artificial passive immunity (2) Natural active immunity

(3) Artificial active immunity (4) Natural passive immunity

12. Which of the following statement is **incorrect** for antibodies

(1) Antibodies present in the colostrum

(2) These are found in blood so response is termed as humoral immune response

(3) These are secreted by T-cells

(4) They have four polypeptide chains

13. Wrong statements about cell mediated immunity is ?

(1) T-supressor cells shows clonal selection and turns into specific clones.

(2) Cell mediated immunity is responsible for graft rejection.

(3) T-helper cells helps B-cells to form antibodies.

(4) T-cytotoxic cells secretes lymphotoxins to kill target cell

14. *Wuchreria* and *Entamoeba histolytica* is respectively transmitted through ?

(1) Culex bite and bird droppings.

(2) Culex bite and sweat.

(3) Culex bite and food or water contaminated with cyst.

(4) Aedes bite and culex bite.

15. Match the following column A with B ?

[A]		[B]	
(i)	BCG	(a)	Live attenuated
(ii)	OPV	(b)	Toxoids + killed vaccine
(iii)	MMR	(c)	Live attenuated Bacteria
(iv)	DPT	(d)	Live combination vaccine

(1) i-a, ii-b, iii-c, iv-d (2) i-a, ii-c, iii-d, iv-b

(3) i-c, ii-a, iii-b, iv-d (4) i-c, ii-a, iii-d, iv-b

16. How many of the followings are infectious diseases ?

(a) AIDS (b) Ring worm (c) Cancer (d) Common cold

(e) Hepatitis (f) Sickle cell anaemia

(1) Two (2) Five (3) Four (4) Three

17. Which of the following is not a character of allergic reaction to dust particles ?

(1) Difficult breathing/wheezing due to bronchioconstriction

(2) Release of histamine by mast cells

(3) Nails appearing bluish due to decreased oxygenation of blood as fluid fills up the alveoli

(4) Production of IgE antibody

18. At which stage of HIV infection does one usually show symptoms of AIDS ?

(1) Within 15 days of sexual contact with an infected person

(2) When the infected retro virus enters host cells

(3) When HIV damages large number of helper T-Lymphocytes

(4) When the viral DNA is produced by reverse transcriptase

19. Which of the following statement is true for Plasmodium ?

(1) Gametocytes develop in liver.

(2) Fever and chills occur due to rupture of liver cells.

(3) Plasmodium reproduce asexually in liver and RBC.

(4) Fertilisation and development takes place in salivary gland of mosquito.

20. Which of the following explanation is/are related with bone marrow and thymus ?

(a) Immature lymphocytes differentiate into antigen sensitive lymphocytes.

(b) Provide micro environment for the development and maturation of T-lymphocyte.

(c) Provide the sites of interaction of lymphocyte with the antigen.

(d) Act as a filter of the body by trapping blood borne microorganisms.

(1) a, b, c & d (2) a & b only (3) a only (4) a, b & c only

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**3**	**1**	**2**	**4**	**3**	**4**	**1**	**1**	**2**	**1**	**3**	**1**	**3**	**4**
Que.	16	17	18	19	20										
Ans.	**3**	**3**	**3**	**3**	**2**										

BIOTECHNOLOGY : PRINCIPLE AND PROCESSES

1. In a gene transfer β-galactosidase gene was used as selectable marker. Host cells were cultured in presence of chromogenic substrate. The scientist was able to identify the recombinant host cells because colonies of these cells were:

(1) Colourless (2) Blue (3) Red (4) Smooth and shinning

2. Which of the following step is not always involved in basic steps in genetically modifying an organism–

(1) Identification of DNA with desirable gene

(2) Introduction of the identified DNA into the host

(3) Amplification of DNA by using PCR

(4) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny

3. Read the following sentences :-

(a) In elution, the separated bands of DNA are cut out from agarose gel & extracted from the gel piece

(b) Wine and beer are produced without distillation.

(c) The downstream processing and quality control testing vary from product to product

(d) Competent bacterial cells cannot take up the plasmid

(1) (a) and (d) are correct

(2) (a, b) and (c) are incorrect

(3) Except (d), all are correct

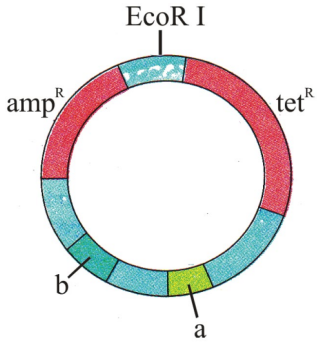
(4) Only (a) is correct

4. Find the correct match :

	Column-I		**Column-II**
(a)	Army worms	(1)	Dipteran
(b)	Flies	(2)	lepidopteran
(c)	Beetles	(3)	Coelopteran
(d)	Mosquitoes		
(e)	Tobacco budworm		

	a	b	c	d	e
(1)	1	3	2	1	3
(2)	2	1	3	1	1
(3)	3	2	1	3	2
(4)	2	1	3	1	2

5. Given below diagram showing cloning vector pBR-322. Mark the incorrect statement :-



(1) Label 'a' represent ROP site, which codes for the proteins involved in the replication of plasmid

(2) Bam HI site is located in amp^R region

(3) Label 'b' repersent 'ori' site

(4) Sal I site is located in tet^R region

6. Which of the following is not a part of biotechnology :-

(1) Gene therapy

(2) Invitro fertilization leading to a test tube baby

(3) Developing of DNA vaccine

(4) Biomagnification of pesticide

7. For transformation with recombinant DNA, the bacterial cells, must first be made 'competent' which means?

(1) Should increase, their metabolic reactions

(2) Should decrease, their metabolic reactions

(3) Increased efficiency with which DNA enters in the bacterium

(4) Ability to divide fast

8. Recombinant DNA technology involves several steps in specific sequence. What is the correct sequence for r-DNA technology ?

a $\rightarrow$ Down stream processing.

b $\rightarrow$ Cutting of DNA at specific location.

c $\rightarrow$ Obtaining the foreign gene product.

d $\rightarrow$ Isolation of the genetic material.

e $\rightarrow$ Insertion of r-DNA into the host cell.

(1) b $\rightarrow$ d $\rightarrow$ e $\rightarrow$ a $\rightarrow$ c

(2) d $\rightarrow$ b $\rightarrow$ e $\rightarrow$ c $\rightarrow$ a

(3) d $\rightarrow$ e $\rightarrow$ b $\rightarrow$ a $\rightarrow$ c

(4) e $\rightarrow$ d $\rightarrow$ a $\rightarrow$ c $\rightarrow$ b

9. PBR^{322} vector is cut by pst - I enzyme and desired gene is ligated in vector at the recognition site of pst I and recombinant DNA is formed, which is transfered into the host cell.

Select the correct statement :-

(1) Recombinant cells will grow in ampicillin containing medium

(2) Non transformant cells will grow in tetracycline containing medium

(3) Recombinant cells will grow in tetracycline containing medium

(4) Non transformant cells will grow in ampicillin containing medium

10. Read the following statements regarding the features of cloning vectors.

(i) The origin of replication is also responsible for controlling the copy number of the linked DNA.

(ii) In addition to 'Ori ; the vector requires a selectable marker, which helps in identifying and eliminating non-transformants and selectively permitting the growth of the transformants.

(iii) The vector needs to have very few preferably single, recognition sites for the commonly used restriction enzymes.

(iv) Presence of more than one recognition sites within the vector will generate several fragments, which will facilitate the gene cloning.

How many of the above statements are correct?

(1) 1 (2) 2 (3) 3 (4) 4

ANSWER KEY										
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	**1**	**3**	**3**	**4**	**2**	**4**	**3**	**2**	**3**	**3**

BIOTECHNOLOGY AND ITS APPLICATION AND MICROBES IN HUMAN WELFARE

1. In anaerobic sludge digester, during digestion, bacteria produce a mixture of gases such as-

(1) CH_4, H_2S and CO_2 (2) CH_4 and H_2S only

(3) H_2S and CO_2 only (4) CH_4 and CO_2 only

2. **Statement-1 :-** In paddy fields, cyanobacteria serve as an important biofertiliser.

Statement-2 :- The Ladybird beetles is useful to get rid of mosquitoes.

(1) Both statement (1) and (2) are correct

(2) Both statement (1) and (2) are incorrect

(3) Only statement (1) is correct

(4) Only statement (2) is correct

3. Read the given statements and select the correct option.

Statement A : Ministry of Environment and Forest has initiated Ganga Action Plan.

Statement B : Effluent from secondary treatment cannot be released into water bodies.

(1) Only A is correct

(2) Only B is correct

(3) Both A & B are correct

(4) Both A & B are incorrect

4. The technology of biogas production in India has been developed by the effort of :-

(A) Khadi and Village Industries Commision (KVIC)

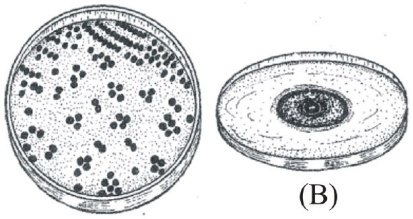
(B) Indian Institute of Technology (IIT)

(C) Indian Agricultural Research Institute (IARI)

(D) Indian Council of Medical Reasearch (ICMR)

(1) A,C,D (2) C,D (3) A,C (4) A,B

5. Refer to the given figure and select the correct match.



(A) (B)

(1) A-Fungal colony, B-Bacterial colony

(2) A-Viral colony, B-Bacterial colony

(3) A-Bacterial colony, B-Viral colony

(4) A-Bacterial colony, B-Fungal colony

6. **Statement-1 :-** Biopiracy is the term used to refer to the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment.

Statement-2 :- In 1983, an Indian company got patent rights on Basmati rice through the US Patent and Trademark Office.

(1) Both Statement-1 and Statement-2 are correct.

(2) Statement-1 is correct but Statement-2 is incorrect.

(3) Statement-1 is incorrect but Statement-2 is correct.

(4) Both Statement-1 and Statement-2 are incorrect.

7. **Statement-A :** The Indian Parliament has recently cleared the fifth amendment of the Indian Patents Bill in 1970.

Statement-B : Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans.

(1) Both A and B are correct (2) Only A is correct

(3) Only B is correct (4) Both A and B are incorrect

8. Select the incorrect match –

(1) Transgenic mice - Polio vaccine

(2) Rosie cow - α lactalbumin gene

(3) ssDNA/RNA probe - Gene therapy

(4) PCR - Molecular diagnosis

9. Which of the following statement is incorrect?

(1) Transgenic mice are used to test safety of polio vaccine

(2) Transgenic animal are use to study toxicity of drugs

(3) First transgenic Cow, Rosie produces human protein α-1 antitrypsin

(4) Transgenic animals are used to study gene regulation and their effect on normal body function

10. **Assertion :** Most of the industrialised nations are rich financially but poor in biodiversity & traditional knowledge.

Reason : In contrast the developing & the underdeveloped world is rich in biodiversity & traditional knowledge related to bio resources.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

11. "The activated sludge pumped into large tanks. Here, other kinds of bacteria, which grow anerobically digest the bacteria and the fungi in the sludge"

This process of sewage treatment performed in which tank ?

(1) Secondary settling tank (2) Aeration tank

(3) Primary settling tank (4) Anaerobic sludge digesters

12. **Statement-I :** The large holes in 'swiss-cheese' are due to production of a large amount of CO_2 by bacterium named *Propionibacterium sharmanii*.

Statement-II : The 'Roquefort cheese' are ripened by growing a specific bacteria on them, which gives them a particular flavour.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

13. **Statement-I :-** Statins act by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Statement-II :- Statins are commercially used as blood cholesterol lowering agent.

(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Statement-I is correct & Statement-II is incorrect.

(4) Statement-I is incorrect & Statement-II is correct.

14. Consider the following statements.

I. Yeast used in making bread and beverages is a prokaryotic organism.

II. Streptokinase is produced by *Streptococcus* and modified by genetic engineering is used as a clot buster.

III. Lipases are added in detergent for removing oily stains from laundry.

IV. With reference to human beings, antibiotics are 'prolife'.

How many of the statement given above are incorrect ?

(1) One (2) Two (3) Three (4) Four

15. Read the following statement and choose the correct statement.

(A) Baculovirus is an biofertilizer

(B) *Glomus* provide phosphorus to plants

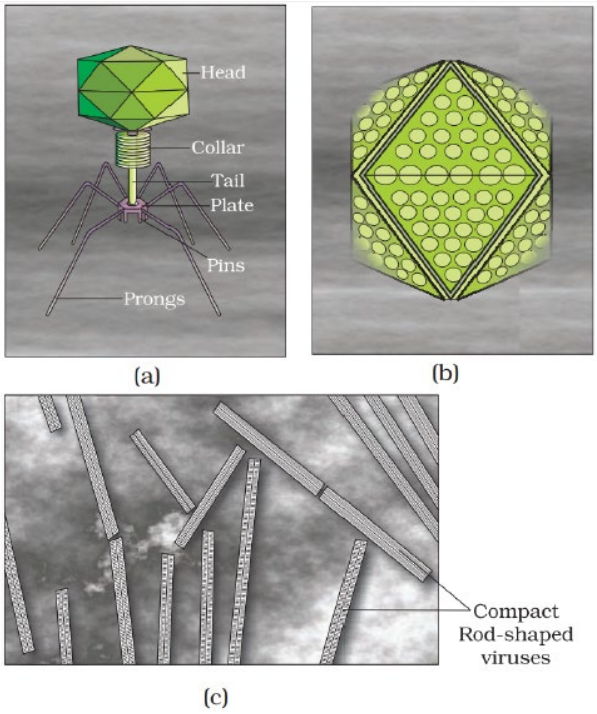
(C) "Cyclosporin-A" produced by the fungus have been commercialised as blood-cholesterol lowering agents

(D) Antibiotics are chemical substance which are produced by some microbes and can kill the growth of other microbes

(1) A, B and D (2) B, C and D

(3) A, C and D (4) B and D

16. Identify the microbes in the given diagrams and select the correct option :



(1) a is adenovirus

(2) c is tobacco mosaic virus

(3) b is bacteriophage

(4) c is adenovirus

17. **Statement-I :-** At present 30 recombinant therapeutic have been marketed for human use in India.

Statement-II :- Recombinant therapeutic do not induce unwanted immunological response as is common in case of similar product isolated from non human source.

Choose the correct answer from option given below :

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Statement-I is correct & Statement-II is incorrect.

(4) Statement-I is incorrect & Statement-II is correct.

18. **Statement-I :-** Genetic modification has made crops tolerant to abiotic stress.

Statement-II :- A nematode *Meloidegyne incognitia* infects the leaves of tobacco plant.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

19. Statements related to human Insulin are given below.

Which statement(s) is/are correct about genetically engineered Insulin?

(a) Pro-hormone insulin contains an extra stretch of C-peptide

(b) A-peptide and B-peptide chains of insulin were produced separately in E.coli, extracted and combined by creating disulphide bond between them.

(c) Insulin used for treating diabetes was extracted from cattles and pigs.

(d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.

(e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below :

(1) (b) only (2) (c) and (d) only

(3) (c), (d) and (e) only (4) (a),(b) and (d) only

20. **Assertion :** The first clinical gene therapy was given in 1990.

Reason : Using conventional methods of diagnosis early detection is not possible.

(1) Both assertion & reason are true & the reasonis a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**3**	**1**	**3**	**4**	**2**	**3**	**3**	**3**	**2**	**4**	**2**	**1**	**1**	**4**
Que.	16	17	18	19	20										
Ans.	**2**	**4**	**3**	**1**	**2**										

ECOLOGY

1. Read statements regarding population interactions:

(A) Biological control methods adopted in agricultural pest control are based on predation.

(B) In the rocky intertidal areas of American pacific coast starfish pisaster is an important predator.

(C) Some species of frog & insect show camouflage

(D) Host specific parasite and host show co-evolution

How many of above statements are **correct** ?

(1) Statement A, B, C, D

(2) Only statement D

(3) Only statement C, D

(4) Only A, B and D

2. **Statement - I :** Any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time.

Statement - II : In nature, a given habitat has limited resource to support a maximum possible number of individuals is called carrying capacity.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Statement-I & Statement-II both are incorrect.

3. 'r' or 'k' selected species theory is concerned with :-

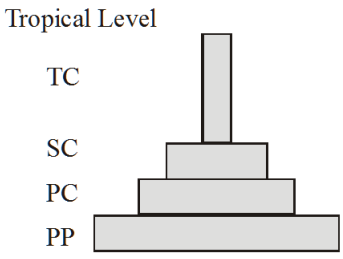
(1) Factors affecting rate of decomposition.

(2) Physiological and behavioural responses of species to cope with variations in their environment.

(3) Positive and negative interspecific population interactions

(4) Evolution of species towards the most efficient reproductive strategy

4. Given pyramid can observed as :-



(1) Pyramid of number without any exception

(2) Pyramid of energy without any exception

(3) Pyramid of biomass without any exception

(4) Both (1) and (3)

5. **Assertion :** In most ecosystems, all the pyramids, of number, of energy and of biomass are upright.

Reason : In a food chain usually producers are more in number and biomass than the herbivores and herbivores are more is number and biomass than the carnivores. Also energy at a lower trophic level is always more than at a higher level.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

6. Find out the correct match from the following table :-

	Column-I	**Column-II**	**Column-III**
(i)	Biodiversity hot spot	Accelerated habitat loss	High biodiversity regions
(ii)	Sacred grove	Last refuges for a large number of rare & threatened species	Khasi & Jaintia Hills in Meghalaya
(iii)	Botanical garden	Off site conservation	Plants are taken out from natural habitat for protection

(1) Only (i) (2) Only (ii) (3) Only (i) & (ii) (4) (i), (ii) & (iii)

7. **Assertion :** Extinctions across taxa are not random.

Reason : Careful analysis of records shows that some groups like amphibians appear to be more vulnerable to extinction.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

8. **Statement-I :** David Tilman found that plots with more species showed more year-to-year variation in total biomass.

Statement-II : Tilman also showed that in his experiments increased diversity contributed to lower productivity due to intense competition.

(1) Statement-I & Statement-II both are correct. (2) Statement-I & Statement-II both are incorrect.

(3) Only Statement-I is correct. (4) Only Statement-II is correct.

9. In species area relationship, if log A = 4, Z = 0.3 and log C = 0.8, find the value of log 'S' ?

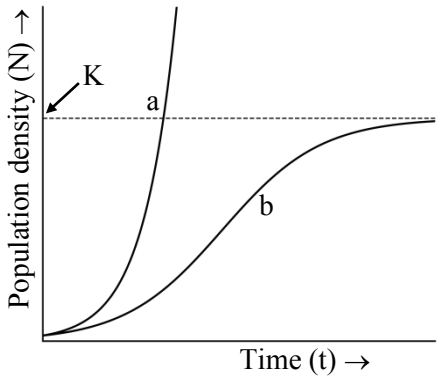
(1) 3.76 (2) 100 (3) 4.24 (4) 2

10. In India we find mangoes with different flavours, colours, fibre content, sugar and even shelf-life. The large variation is on account of :-

(1) Genetic diversity (2) Species diversity

(3) Ecological diversity (4) Biomes diversity

11. Select the correct option w.r.t. given population growth curve :



(1) a = resources are not limiting the growth; $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

(2) b = resources are limiting the growth; $\frac{dN}{dt} = rN$

(3) a = growth is exponential; $N_t = N_0 e^{rt}$

(4) b - growth is logistic ; $\frac{dN}{dt} rN\left(\frac{N-K}{K}\right)$

12. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as Reason R.

Assertion (A) : Predators help in maintaining species diversity in a community.

Reason (R) : Predators reduces the intensity of competition among competing prey species.

In the light of the above statements, choose the correct answer from the options given below :-

(1) Both A and R are true but R is NOT the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A.

13. Which are correct from following ?

(A) Population ecology links ecology to population genetics and evolution.

(B) Birth rate and death rate are features of an organism.

(C) If a new habitat is just being colonised, immigration may contribute more significantly to population growth than birth rate.

(D) The exponential growth model is considered a more realistic one.

(1) A & B are correct (2) B & C are correct

(3) C & D are correct (4) A and C are correct

14. Which of the following statement is incorrect?

(1) Photosynthetically Active Radiation(PAR) is less than 50% of Incident Solar Radiation (ISR)

(2) Decomposition rate is not affected by climatic factors.

(3) Atmospheric input of phosphorus through rainfall are much smaller than carbon input

(4) According to Allen's rule mammals from colder climate generally have shorter ears and limbs.

15. Tree → insect → bird(small) → large bird.

This is a type of food chain. What will be the shape of pyramids :-

(a) Pyramids of number.

(b) Pyramid of energy.

(1) (a) inverted, (b) upright (2) (a) spindle, (b) upright

(3) (a) spindle, (b) spindle (4) (a) upright, (b) upright

16. Select the incorrect statement about biodiversity :-

(1) Increases from high to low latitudes

(2) Decrease of biodiversity promote niche specilization

(3) Habitat destruction or loss is the primary cause for the loss of biodiversity

(4) Whooping crane and blue whale are more susceptible to extinction due to fixed habitat and migratory routes

17. Regarding the distribution of species waiting to be discovered, which statement is incorrect?

(1) Many taxonomic groups have more complete inventories in tropical countries compared to temperate regions

(2) Estimates regarding undiscovered species vary widely, and many are only educated guesses.

(3) The overwhelming majority of undiscovered species are believed to be in tropical regions

(4) Given by Robert May applying proportion to India diversity we estimate that there are probably more than 1,00,000 plants species and more than 3,00,000 animal species yet to be discovered.

18. Identify the **correct** combination :

	(a)	**(b)**	**(c)**
(1)	Sea anemone	Clown fish	Commensalism
(2)	Whale	*Barnacles*	Mutualism
(3)	Fig tree	Wasp	Commensalism
(4)	Cuckoo	Crow	Mutualism

19. **Assertion :** *Cuscuta* is a parasitic plant that commonly found growing on hedge plants.

Reason : *Cuscuta* has lost its chlorophyll and leaves in the course of evolution. It derives its nutrition from the host plant.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

20. In a month of January Siberian Cranes migrate from Russia to India for breeding. A survey was done-

· Till December total population of Siberian Cranes = 1200

· Birth Rate = 400

· Mortality Rate = 200

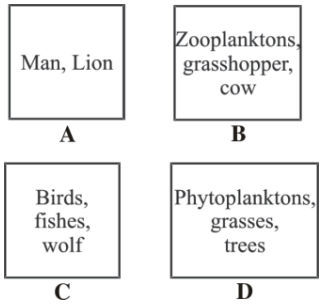
· Number of Cranes immigrated = 600

· Number of cranes emigrated = 300

Calculate the total population :-

(1) 1500 (2) 1000 (3) 2000 (4) 1700

21.



Which of the following are **incorrect** for the boxes given above ?

I. Members of box 'D' are more adversely affected by competition, than the members of box 'A'.

II. Any member of any box given can not be found linked with DFC (Detritus food chain)

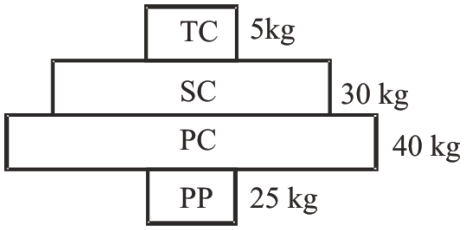
III. Members of box A are top consumers and they are given top positions in ecological pyramids based on GFC (Grazing food chain).

IV. Members of box B and C can be present in a GFC as trophic level first and second respectively.

V. Members of box D are primary producers in GFC.

(1) I and III (2) II and IV (3) II and V (4) I, III and V

22.



Given above is an imaginary pyramid of biomass, what could be the possibilities about certain organisms at different trophic levels ?

(1) level PP is tree and level SC is insectivorous birds

(2) Level PP is phytoplankton & level SC is zooplankton

(3) Level PP is grassland and level SC is goat

(4) Level PP is phytoplankton and level TC is large fish

23. If a food chain is composed of only 3 trophic levels, then how much energy will be transfered to the third trophic level ?

(1) 1% of 1st trophic level (2) 20% of 2nd trophic level

(3) 10% of 1st trophic level (4) 0.1% of 2nd trophic level

24. **Assertion :** For many taxonomic groups, species inventories are more complete in temperate than in tropical countries.

Reason : Conventional taxonomic methods are not suitable for identifying species in tropics.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

25. **Statement-I :** A single species might show high diversity at the genetic level over its distributional range.

Statement-II : The diversity of plants and animals is uniform throughout the world.

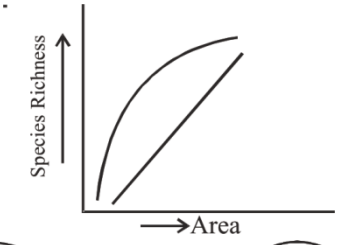
(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct but Statement-II is incorrect.

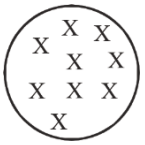
(3) Statement-I is incorrect but Statement-II is correct.

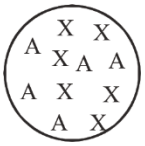
(4) Statement-I & Statement-II both are incorrect.

26. In the below graph relation between species richness and area for a wide variety of taxa is shown. On a logarithmic scale, the relationship shown by straight line, In which area slope of line to be much steeper ? Here Area (A) = 1 km^2



(1) 

(2) 

(3) 

(4) 

27. Which of the following are generally considered to be the major reasons for biodiversity conservation ?

(i) Economic benefits

(ii) Ethical reasons

(iii) Political advantages

(iv) For scientific study

(v) Ecological benefits

(1) (i), (ii), (v)

(2) (i), (iii), (v)

(3) Only (ii), (v)

(4) Only (ii), (iii)

28. How many characteristics in the list given below are structural ?

Dominance, Energy use efficiency, Species diversity, Species composition, Stratification, Organism size, Nutrient conservation, Food web complexity, physiognomy.

(1) Three

(2) Four

(3) Five

(4) Six

29. Which one of the following is most appropriately defined -

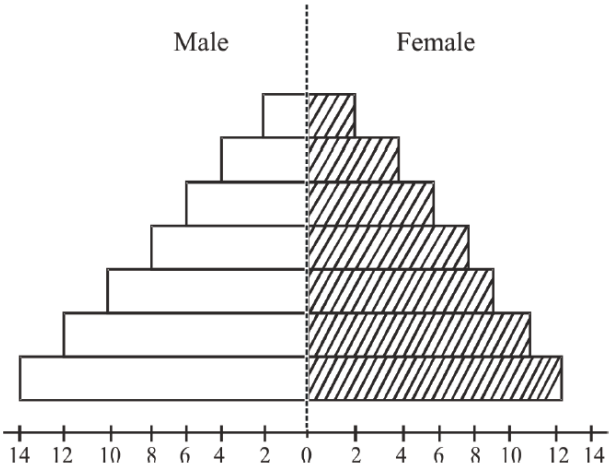
(1) Host is an organism which provides food to another organism

(2) Amensalism is a relationship in which one species is benefited whereas the other is unaffected

(3) Predator is an organism that catches and kills other organism for food

(4) Parasite is an organism which always lives inside the body of other organism and may kill it

30. Study the given age pyramid diagram & identify appropriate option :-



(1) There is low sex ratio in this population in higher age group.

(2) It represents a stable population.

(3) There is low sex ratio in this population in children (pre-reproductive) age group.

(4) The pyramid shows low IMR.

31. If a pond has 40 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 48, what will be the birth rate ?

(1) 0.6 offspring per lotus per year

(2) 0.5 offspring per lotus per year

(3) 0.2 offspring per lotus per year

(4) 0.1 offspring per lotus per year

32. Why is net primary production (NPP) a more useful measurement to an ecosystem ecologist than gross primary productivity (GPP)?

(1) NPP is same as the standing crop

(2) NPP shows the rate at which the standing crop is utilized by consumers

(3) NPP represents the stored chemical energy that is available to consumers in the ecosystem.

(4) NPP can be expressed in terms of carbon fixed by photosynthesis for an entire ecosystem

33. Consider the following statements about decompositions.

(a) Decomposition occurs only in aerobic conditions.

(b) Decomposition releases nutrients back into the ecosystem.

(c) Decomposition is influenced by temperature and moisture.

How many statements are correct ?

(1) Only a

(2) Only b

(3) b and c

(4) a, b, and c

34. Select the incorrect statement -

(1) Protection of hot spot can induce ongoing mass extinction by almost 30%

(2) India covers about 2.4% of world land area and shares 8.1% of total global diversity.

(3) Exploring molecular, genetic and species level for products of economic importance is called Bio-prospecting.

(4) The world summit on sustainable development was held in 2002.

35. David Tilman's long-term ecosystem experiments using outdoor plots showed that:-

(1) Increased diversity contributed to lower productivity

(2) Increased diversity contributed to higher productivity

(3) Increased diversity have no impact on productivity

(4) Increased diversity have unpredictable impact on productivity

ANSWER KEY															
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**1**	**4**	**2**	**1**	**4**	**1**	**2**	**4**	**1**	**3**	**4**	**4**	**2**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**1**	**1**	**1**	**4**	**2**	**4**	**1**	**3**	**2**	**4**	**1**	**4**	**3**	**3**
Que.	31	32	33	34	35										
Ans.	**3**	**3**	**3**	**1**	**2**										